MILITARY STANDARD

SOLDERING TECHNOLOGY, HIGH QUALITY/HIGH RELIABILITY

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2. MAKE THE FOLLOWING PEN AND INK CHANGES:

a. Page 5, paragraph 4.11.1.3:

(1) Line 1: Insert "and tools" after "areas."

(2) Line 2: Change "flux," to "flux residue,".

b. Page 7, paragraph 4.11.3:

(1) Line 4: Delete "inject" after "not" and substitute "compromise functional integrity by injecting."

(2) Line 5: Add "being" before "cleaned." Add period after "cleaned" and delete "to the extent ..... is compromised."

(3) Lines 6, 11 and 12: Delete "earth."

(4) Line 7: Change "20 ohms" to "2.0 ohms."

(5) Delete entire last sentence, "Magnetic fields ..... items processed."

c. Page 8, paragraph 4.11.3.1:

(1) Line 10: Add "RMS" after "millivolts."

(2) Lines 10 and 11: Delete "voltage sensitive devices are soldered." after "used when" and substitute "soldering."

d. Page 8, paragraph 4.11.3.5: Line 2 and line 1 of subparagraph a: Delete "assembly and."

e. Page 19, paragraph 4.19.1:

(1) Line 7: Insert "(actual measurement not required except for referee purposes)" after "hole depth (d)."

(2) Lines 16 and 17: Delete "whether or not wetting to both the lead and terminals areas is acceptable" and replace with "(actual measurement not required except for referee purpose)."

f. Page 21. Add the following new paragraph:

4.21 Markings and designators. Markings and designators shall be in accordance with Requirement 67 of MIL-STD-454.

g. Page 22, paragraph 5.1.2, line 5: Change ":4.11.3.4" to "5.2.2.2."

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3. RETAIN THIS NOTICE AND INSERT BEFORE TABLE OF CONTENTS.

4. Holders of DOD-STD-2000-1B will verify that page changes and additions indicated above have been entered. This notice page will be retained as a check sheet. This issuance, together with appended pages, is a separate publication. Each notice is to be retained by stocking points until the military standard is completely revised or canceled.

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Preparing Activity:
Navy - AS
(Project No. SOLD-0024)

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FOREWORD

1. Electrical and electronic equipment and systems used in military applications are continuously becoming more diverse, complex, and sophisticated yet smaller, lighter, denser in internal packaging, sturdier with respect to environment, and more reliable with respect to both function and service life. Such changes in technological posture mandate both definition and standardization of connections and wiring in military electrical and electronic equipment.

2. This standard provides for communication between designers, fabricating personnel and inspectors. The uniformity of soldering materials and processes applies to manual or automated procedures. This standard is intended for use by any of the many organizations engaged in the production of military electronic equipment. Criteria of this standard are not directed to end item products per se but are instead directed to part and component solder connections and wiring essential to the said end item products and should be implemented in conjunction with appropriate documents.

3. This standard addresses that spectrum of solder connections normally defined as standard and miniature, and provides technical criteria essential to electrical connections and wiring within equipment fabricated by or for the Department of Defense. Requirements for microminiature soldering applicable to connections in thin and thick film microelectronic assemblies are not specifically defined but are imposed through callout of MIL-M-38510.

4. It is recognized that solder processes other than specified herein exist or may exist in the future but it is intended that the requirements of this standard be extrapolated to such noncovered processes insofar as practicable.

5. DOD-STD-2000-1, DOD-STD-2000-2, and DOD-STD-2000-3 are standards concerning the requirements for High Quality/High Reliability soldered electrical and electronic connections. They are intended to be invoked only for this level of "Quality". The three documents are also intended to be used as a set. The requirements in each consider and complement the others.

6. These three documents are not to be confused with DOD-STD-2000-4 General Purpose Soldering Requirements for Electrical and Electronic Equipment. DOD-STD-2000-4 covers general purpose soldering only, and will be invoked independently in situations which do not require soldering at the level covered by DOD-STD-2000-1, DOD-STD-2000-2, and DOD-STD-2000-3.

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1. SCOPE

1.1 Applicability. This standard is applicable to soldering processes for high quality/high reliability electrical and electronic connections on guided missiles, aircraft, avionics systems, communication equipment, satellites, shipboard weapons systems, ammunition and weapon systems, ground vehicle equipment, and program critical ground support equipment.

1.2 Classification. The soldering processes specified herein are of the following classes:

Class 1. Manual Soldering (5.1)
Class 2. Automated Soldering (5.2)

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this standard to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-E-760	Ethyl Alcohol (Ethanol) Denatured Alcohol, Proprietary Solvents and Special Industrial Solvents
O-M-232	Methanol (Methyl Alcohol)
QQ-S-571	Solder, Tin Alloy, Lead-Tin Alloy and Lead Alloy
TT-B-848	Butyl Alcohol, Secondary, For Use in Organic Coatings
TT-I-735	Isopropyl Alcohol

MILITARY

MIL-I-7444	Insulation Sleeving, Electrical Flexible
MIL-F-14256	Flux, Soldering, Liquid (Rosin Base)
MIL-I-22076	Insulation Tubing, Electrical Nonrigid, Vinyl, Very Low Temperature Grade
MIL-E-22118	Enamel, Electrical, Insulating
MIL-I-23053	Insulation Sleeving, Electrical, Heat Shrinkable, General Specification for
MIL-M-38510	Microcircuit, General Specification for

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MIL-C-45224	Cable and Harness Assemblies, Electrical, Missile System, General Specification for
MIL-I-46058	Insulating Compound, Electrical for Coating Printed Circuit Assemblies
MIL-P-50884	Printed Wiring, Flexible and Rigid-Flex
MIL-P-55110	Printed Wiring Boards
MIL-C-81302	Cleaning Compound, Solvent, Trichlorotrifluoroethane
MIL-T-81533	Trichloroethane 1, 1, 1 (Methyl Chloroform) Inhibited, Vapor Degreasing
MIL-S-83519	Splice, Shield Termination, Solder Style, Insulation, Heat Shrinkable, Environment Resistant, General Specification for
MIL-C-85447	Cleaning Compounds, Electrical and Electronic Components

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-1389	Design Requirements for Standard Electronic Modules
DOD-STD-1686	Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies, and Equipment (Excluding Electrically-Initiated Explosive Devices) (Metric)
DOD-STD-1866	Soldering Process, General (Non-Electrical) (Metric)
DOD-STD-2000-2	Part and Component Mounting for High Quality/High Reliability Soldered Electrical and Electronic Assemblies
DOD-STD-2000-3	Criteria for High Quality/High Reliability Soldering Technology

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MIL-STD-2118 Flexible and Rigid-Flex Printed Wiring for Electronic Equipment, Design Requirements for

FEDERAL

FED-STD-376 Preferred Metric Units for General Use by the Federal Government

HANDBOOKS

MILITARY

MIL-HDBK-141 Optical Design

DOD-HDBK-263 Electrostatic Discharge Control Handbook for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding Electrically Initiated Explosive Devices) Metric

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Government contracting officer).

2.2 Other publications. The following document(s) form a part of this standard to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue in the DODISS and the supplement thereto, if applicable.

ANSI/IPC-T-50 Terms and Definitions for Interconnecting and Packaging Electronic Circuits

IPC-SM-840 Printed Board, Permanent, Polymer Coating (Solder Mask) for Qualification and Performance of

(Application for copies should be addressed to the Institute for Interconnecting and Packaging Electronic Circuits, 7380 North Lincoln Avenue, Lincolnwood, IL 60646.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 1193 Reagent Water

ASTM D 3295 Polytetrafluoroethylene Tubing

ASTM E 8 Metallic Materials, Tension Testing of

ASTM E 53 Chemical Analysis of Copper (Electrolytic Determination of Copper), Method for

ASTM E 345 Tension Testing of Metallic Foil, Methods of

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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CFR, Title 29, Part 1900 to 1919, Chapter XVII — Code of Federal Regulations, Occupational Safety and Heath Administration, Department of Labor

(Application for copies should be addressed to the American Conference of Industrial Hygienists, 6500 Glenway, Building D-7, Cincinnati, OH 45211.)

Industrial Ventilations, Manual of Recommended Practices

(Application for copies should be addressed to: Committee of Industrial Ventilation, P.O. Box 16153, Lansing, MI 48902.)

3. DEFINITIONS

3.1 Terms and definitions. The definitions applicable to this standard shall be in accordance with ANSI/IPC-T-50.

4. GENERAL REQUIREMENTS

4.1 Conflict. In the event of any conflict between the requirements of this standard and the applicable assembly drawing(s), differences shall be referred to the Government Contracting Officer or his designated technical activity and a request for deviation with supporting data shall be submitted for approval prior to proceeding. Upon such approval, the provisions shall be officially documented (by notice of revision or equivalent) on the assembly drawing(s) which shall then govern.

4.2 Requirements flowdown. The requirements of the DOD-STD-2000 series of standards shall be imposed by each contractor on all applicable subcontractors requirements and purchase orders. The contractor shall not impose or allow any variation from these standards on subcontractors or purchase orders other than those which have been approved by the Government Contracting Officer for the applicable prime contract.

4.2.1 Existing designs. The requirements of this standard shall not constitute the sole cause for redesign if the design was approved prior to 1 January 1987. However, when drawings for such existing designs undergo revision for any reason, they shall be reviewed and changes made that allow for maximum practical compliance with the requirements of this standard.

4.3 Visual aids. Line drawings, illustrations and photographs depicted either herein, DOD-STD-2000-2, or DOD-STD-2000-3, and any additional provisions approved by the procuring Government Contracting Officer are provided as aids for determining compliance with the written requirements of this standard and shall not take precedence over the written requirements.

4.4 Visual inspection. One hundred percent visual inspection of all soldered connections and assemblies shall be performed using the magnification aids of paragraph 4.11.3.5. The soldered connections and assemblies shall conform to the requirements specified herein, DOD-STD-2000-2 and -3.

4.5 Hybrid microelectronic modules and assemblies. Hybrid microelectronic processing, modules, and assemblies shall be in accordance with MIL-M-38510.

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TABLE I. Solvents.

Solvent (see 4.16.4.1 and 4.16.4.2)	Specification
Ethyl Alcohol	O-E-760, Types III, IV or V
Isopropyl Alcohol	TT-I-735
Methyl Alcohol (see 4.16.4.5)	O-M-232, Grade A
Butyl Alcohol, Secondary (for Use in Organic Coatings (see 4.16.4.5)	TT-B-848
1, 1, 1-Trichloroethane (see NOTE)	MIL-T-81533
Trichlorotrifluoroethane (see 4.16.4.3 and 4.16.4.4)	MIL-C-81302
Trichlorotrifluoroethane (see 4.16.4.3 and 4.16.4.4)	MIL-C-85447, Type II
Solvent Petroleum Distillate (Stoddard) (see 4.16.4.6)	Use Appendix B

TABLE II. Cleaners.

Cleaners	Specification/Note
Reagent water (Type II)	ASTM D 1193
Detergent cleaners and saponifiers	As approved by the Government Contracting Officer prior to use

NOTE: 1, 1, 1-Trichloroethane may attack plastics commonly used in electronic assemblies, including acrylics, polycarbonates and may also attack rubber, neoprene and silicone; consequently, use of this solvent should be restricted or carefully controlled. (Do not mix with water.)

4.16.4.1 Mixtures of the approved solvents may be used. Solvents used shall be selected for compatibility with the materials being cleaned and for the capability of removing both ionic and nonionic contamination.

4.16.4.2 To prevent contamination of parts and to provide for personnel safety, areas for cleaning parts and areas where toxic or volatile vapors are generated shall have an exhaust system to remove the vapors as they are generated.

4.16.4.3 Solvents in which a nitromethane stabilizer is blended with trichlorotrifluoroethane conforming to either MIL-C-81302 or MIL-C-85447 (Type II) are also acceptable provided such mixtures are preblended by the supplier.

4.16.4.4 Trichlorotrifluoroethane blends which include ketones or halocarbons other than chlorofluorocarbons defined in MIL-C-81302 or MIL-C-85447 (Type II) are unacceptable. Mixtures of trichlorotrifluoroethane and water shall not be used.

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4.16.4.5 Methyl alcohol and secondary butyl alcohol shall be used only when purchased as a constituent of an already blended solvent. Pure methyl alcohol or secondary butyl alcohol shall not be used alone as a solvent.

4.16.4.6 Solvent in accordance with Appendix B shall be used only as a constituent of an already blended solvent. The pure solvent shall not be used unblended. The blended solvent shall not be used as a final cleaner.

4.16.5 Tubing. Polytetrafluoroethylene tubing shall conform to ASTM-D-3295.

4.16.6 Heat shrinkable tubing. Heat shrinkable tubing shall conform to MIL-I-23053 except that use of polyvinylchloride tubing is prohibited.

4.16.7 Extruded vinyl plastic tubing. Extruded vinyl plastic tubing shall conform to MIL-I-7444, type I, or MIL-I-22076.

4.16.8 Rigid printed wiring boards. Rigid printed wiring boards (PWB) shall be designed to MIL-STD-275 and fabricated to MIL-P-55110 except as modified herein (DOD-STD-2000-1, -2, and -3) (see 4.18.10).

4.16.8.1 Copper plating. Copper plating of the plated-through-holes shall be accomplished utilizing solutions that will not affect quality of plated-through-holes before, during and after exposure to solder reflow or wave soldering temperature. The copper plating process shall meet the following requirements:

a. Purity: 99.8 minimum (copper sulfate type bath)
99.5 minimum (pyrophosphate type bath)

b. Elongation: 6 percent minimum

c. Tensile strength: 36,000 pounds per square inch (psi) minimum (250 MPa)

4.16.8.2 Finish plate. Immersion-tin or electroless-tin plating shall not be used as a finishing process. As an exception, immersion and electroless-tin plating may be used on special R.F. circuitry applications with prior approval of the Government Contracting Officer.

4.16.8.3 Surface finish (coating). The fuzed tin-lead plating or solder coating shall be continuous. There shall be no dewetting or exposed basis metal in plated-through holes or on the top surface of conductors and terminal areas.

4.16.9 Flexible and rigid-flex printed wiring. Flexible and rigid-flex printed wiring shall be designed to MIL-STD-2118 and fabricated to MIL-P-50884 except as specified herein (DOD-STD-2000-1, DOD-STD-2000-2 and DOD-STD-2000-3) (see 4.18.10).

4.16.9.1 Copper plating. Copper plating of the plated-through holes shall be accomplished utilizing solutions that will not affect quality of plated-through holes before, during and after exposure to solder reflow or wave soldering temperature. The copper plating process shall meet the following requirements:

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a. Purity: 99.8 minimum (copper sulfate type bath)
99.5 minimum (pyrophosphate type bath)

b. Elongation: 6 percent minimum

c. Tensile strength: 36,000 pounds per square inch (psi) minimum (250 MPa)

4.16.9.2 Finish plate. Immersion-tin or electroless-tin plating shall not be used as a finishing process. As an exception, immersion and electroless-tin plating may be used on special R.F. circuitry applications with prior approval of the Government Contracting Officer.

4.16.9.3 Surface finish (coating). The fuzed tin-lead plating or solder coating shall be continuous. There shall be no dewetting or exposed basis metal in plated-through holes or on the top surface of conductors and terminal areas.

4.16.10 Solder masks and localized maskants. Polymer solder mask coatings shall conform to IPC-SM-840, Class 3 and shall be applied only to surfaces which are not coated with solder or other material which will become semi or totally liquid during subsequent assembly operations. Dry film polymer mask shall be applied by a vacuum laminant process. Polymer solder mask coatings and localized maskants shall also be of a material that:

a. Does not degrade the substrate material or printed wiring;

b. Precludes solder flow to the masked area;

c. Is compatible, if left in place, with printed wiring board basis material, solder, printed wiring, and subsequently applied conformal coatings; and

d. Can be readily removed without post-removal residual contamination harmful to board integrity if removal is essential.

4.17 Solderability.

4.17.1 Solderability of leads and terminations. Within 30 days of receipt, component leads, terminations and end caps to be soldered shall be tested to and conform to the solderability tests specified in MIL-STD-202, Method 208 using as a minimum, a sampling plan in accordance with MIL-STD-105, Inspection Level S-2, Acceptable Quality Level (AQL) 1.0 percent defective. If more than 120 days elapse before use, the items shall be retested to and conform to MIL-STD-202, Method 208 or shall be hot solder dipped to provide a fused coating with a minimum thickness of 0.0001 inch (0.0025 mm). Solderability shall be tested after tinning using the same Method less steam aging and the same sampling plan. If the items meet solderability requirements, they may be used immediately or stored for one year without further testing, provided issuance to production is in accordance with a first-in first-out (FIFO) system. If these items are stored more than one year after tinning, solderability testing including steam aging using the same inspection level and AQL, shall be performed prior to use.

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4.17.2 Solderability of boards. The solderability of printed wiring boards shall conform to the requirements of MIL-P-55110 for rigid printed wiring and MIL-P-50884 for flexible and rigid-flex printed wiring. The solderability of printed wiring stored for more than one year shall be reverified in the same manner prior to use in accordance with MIL-STD-105 inspection level S-2, AQL 1.0 percent defective.

4.17.3 Barrier coatings. Components with sintered-metal-on-ceramic terminations shall be utilized in printed wiring assemblies only if a diffusion barrier coating of electrolytically-deposited nickel conforming to QQ-N-290, Class 2 is incorporated between the deposited-and-sintered precious metal and the finish coating. The nickel barrier coating shall be at least 50 microinches thick and the finish coating shall be a gold plating 50-100 microinches thick or a fused tin-lead alloy conforming to Sn63 of QQ-S-571 at least 300 microinches thick.

4.18 Preparation for soldering.

4.18.1 Insulation removal. Insulation shall be removed from wire conductors by one of the following methods, using the tools specified in 4.11.2. After insulation removal, insulation deformation shall not exceed 20 percent of the insulation thickness. The insulation shall not have gouges, ragged edges nor be loose or frayed. Slight discoloration of the insulation from thermal stripping is acceptable. At no time during the soldering process shall insulation that is degraded from the use of solvents or chemical stripping agents be acceptable. Wires or strands shall not be broken, severed, nor show evidence of nicks, cuts, scrapes, stretching, or other observable damage exceeding 5 percent of the original diameter of any wire strand when viewed under 4X magnification. There shall be no birdcaging of the wire strands. No basis metal shall be exposed. Discoloration of the wires or strands that shows evidence of overheating shall be cause for rejection.

4.18.1.1 Thermal. When extruded insulation is removed using a thermal stripping tool, the lay of the wire shall be restored, if disturbed, without using bare finger contact. Thermal type insulation strippers are preferred for wires of size 20 AWG and smaller.

4.18.1.2 Mechanical. When extruded insulation is removed using a mechanical stripping tool, the lay of the wire strands shall be restored, if disturbed, without using bare finger contact. A device that utilizes fiberglass stripping wheels may be used for magnet wire.

4.18.1.3 Chemical. Insulation removal from magnet wire shall be accomplished by the use of chemical stripping agents, in accordance with the manufacturer's recommendation. Chemical stripping agents shall be neutralized or removed prior to soldering.

4.18.1.4 Solder dip. Polyurethane or similar type coatings may be removed by dipping the insulated wire to the required depth into a solder pot (see 4.11.3.4) at the temperature recommended by the manufacturer. Stripping of solder strippable magnet wire (with polyurethane or similar insulation) on the termination of coils and windings wound with such wire may be performed by hot solder application in compliance with the wire manufacturers recommendations.

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4.18.2 Preparation of gold plated areas. Gold plated areas of connectors and component leads shall not be utilized when thickness of the plating is less than 0.000050 inch (.00127 mm). Areas to be soldered on gold plated connectors and component leads, the gold plating of which is in the range of 0.000050 to 0.000100 inch (.00127 to .00254 mm) thickness shall be single dip tinned. Areas to be soldered which have gold plating exceeding 0.000100 inch (.00254 mm) shall be prepared for soldering by implementation of either of the two following methods:

Method 1. Static bath. Immerse the fluxed gold plated lead in solder bath number one for 2 to 5 seconds. Only the portion of the lead subsequently to be soldered need be immersed. Gold contamination level in solder bath number one shall be less than 4 percent. Immerse the fluxed tinned lead in solder bath number two for 2 to 5 seconds. Gold contamination in bath number two shall not exceed the limits specified in table III.

Method 2. Dynamic bath. Immerse the fluxed gold plated lead in a flowing solder bath for 2 to 5 seconds. Only the portion of the lead to be subsequently soldered need be immersed. Gold contamination in the solder bath shall not exceed the limits of column 1 (Preconditioning) of table III.

TABLE III. Solder bath contaminant levels.

Contaminant	Maximum allowable percent of contamination per solder operation		Testing frequency 8 hr. operating days 5/		
	Preconditioning (Lead or wire) tinning 1/	Machine soldering 1/, 2/	Columns		
			A	B	C
Copper	0.75	0.30	15	30	30
Gold	0.50	0.20	15	30	30
Cadmium	0.01	0.005	15	30	60
Zinc	0.008	0.005	15	30	60
Aluminum	0.008	0.006	15	30	60
Antimony 3/	0.20 to 0.50	0.20 to 0.50	15	60	120
Iron	0.02	0.02	15	60	120
Arsenic	0.03	0.03	15	60	120
Bismuth	0.25	0.25	15	60	120
Silver 4/	0.75	0.10	15	60	120
Nickel	0.025	0.01	15	60	120

1/ The tin content of the solder shall be from 59.5 to 63.5 percent tin and tested at the same frequency testing for copper or gold contamination. The balance of the solder shall be lead with one or more of the listed contaminants.

2/ The total of copper, gold, cadmium, zinc and aluminum contaminants shall not exceed 0.4 percent for soldering.

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3/ The minimum amount of antimony present in the solder for either operation (column 1 or column 2) shall be 0.20 percent.
4/ Not applicable for Sn62 solder. Limits to be 1.75 to 2.25 (both operations).
5/ An operating day constitutes any 8 hour period, or any portion thereof, during which the solder is liquified and used.

4.18.2.1 Cup type connections. The inside portion of cup type connections shall be tinned in accordance with a or b below:

a. Steps 1 through 6 below shall be performed if solder cups are not gold plated or are gold plated to a thickness between 0.000050 and 0.000100 inches (0.00127 and 0.00254 mm).

b. Solder cups gold plated in excess of 0.000100 inches (0.00254 mm) shall be tinned in accordance with steps 1 through 7 below.

Step 1: Clean solder cups using a solvent or cleaner specified in 4.16.4.

Step 2: Place sufficient solder in the solder cup to completely fill the solder cup, to the milled lip, after melting.

Step 3: Heat solder cup sufficiently to melt solder and allow all gases and flux to escape (cups should be at approximately a 45 degree angle to prevent entrapment of gases and flux).

Step 4: Remove solder by wicking or extraction device.

Step 5: Inspect inside of cup to ensure complete tinning of solder cup.

Step 6: Place sufficient solder in the solder cup to allow proper fillet after wires are inserted into the cup (see 5.1.8).

Step 7: Repeat steps 3 through 6 one time.

4.18.2.2 Post gold removal processing. Upon conclusion of the gold removal process, the solder coated portion of the component shall conform to the Acceptance Criteria of MIL-STD-202, Method 208. If parts are not soldered into an assembly within 1 year of gold removal, the parts shall be tested and shall meet the solderability requirements of paragraph 4.17.1.

4.18.3 Automatic lead forming devices. Automatic lead forming devices are acceptable. Smooth impression marks (base metal not exposed) resulting from bending tool holding forces shall not be cause for rejection.

4.18.4 Lead bends. The distance between the body of the component or weld and the bent section of a lead shall be in accordance with mounting criteria specified in DOD-STD-2000-2.

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4.18.5 Stress relief. Components with wires or leads terminated at a solder connection shall be mounted to assure stress relief of the component in accordance with DOD-STD-2000-2.

4.18.5.1 Solder in the lead bend radius. Solder fillets shall not extend into the bend radius on any lead bend configuration except for axial-leaded components with a body diameter of 0.125 inch (3.18 mm) or less and with leads formed to the 90-degree bend as shown in figures 2A and 2B. Components with body diameters of 0.125 inch (3.18 mm) or less and lead configurations of figures 2A and 2B may have a solder fillet which extends into the bend radius on one lead only but this solder fillet shall not touch the component body or end seal.

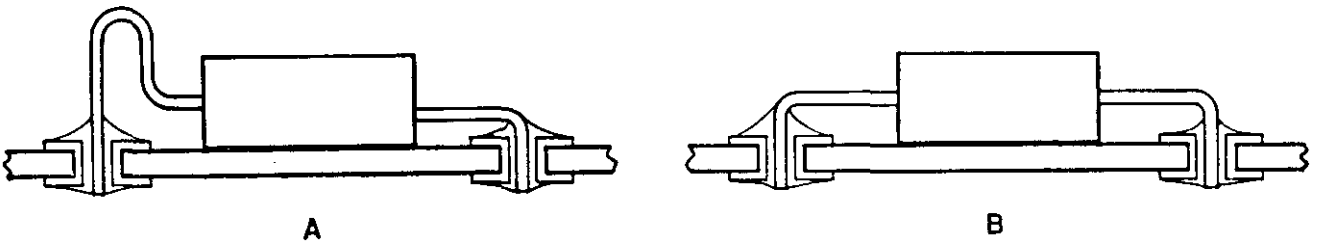


FIGURE 2. Solder in the lead bend radius (see 4.18.5.1).

4.18.6 Lead trimming. Wires and leads shall be cut to their proper length prior to soldering. When leads are cut after insertion in the printed wiring board, the cutting action shall not damage the printed wiring board or components mounted thereon. When automatic lead cutting is performed, a maintenance program shall be implemented to provide assurance that blades or saws are monitored for wear, and that the feed system is calibrated to match the needs of the materials and density of leads being cut. Cutting shall not bend the lead nor leave sharp spurs on lead ends.

4.18.7 Tinning of stranded wire. Prior to the soldering operation, that portion of stranded wire to be soldered to another surface shall be tinned (see figure 3) with the solder penetrating to the inner strands of the wire. Tinning of stranded wire shall not obscure the wire contour of the individual outer strands and the entire stripped end shall be tinned to within 1/16 ± 1/32 inch (1.6 ± 0.8 mm) from the end of the insulation.

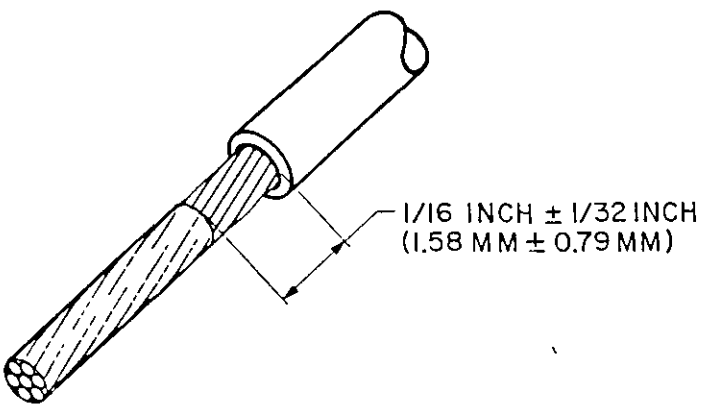


FIGURE 3. In-process step for tinning stranded wire (see 4.18.7).

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4.18.8 Precleaning. Items shall be clean prior to soldering or bonding. The solvents of 4.16.4 shall be used for cleaning.

4.18.9 Handling and storage of parts. Leads and to-be-soldered surfaces of component wires and printed wiring boards shall not be handled with bare hands. If the parts cannot be handled without touching the surface to be soldered, protective devices such as clean nylon or cotton gloves, finger cots, or special tooling shall be used.

CAUTION: Finger cots and nylon gloves should not degrade electrostatic discharge protection as defined in DOD-STD-1686.

NOTE: Printed wiring boards should be oven dried prior to being used in any soldering process to prevent defects such as measles, crazing and delaminations.

4.18.10 Printed wiring board condition. Before components or terminals are mounted on a printed wiring board, the board shall have been examined for conformance to the requirements of 4.16.8, 4.16.9 and 4.17.2. There shall be no evidence of any of the following defects:

a. Scratches which expose basis metal.

b. Separation of the conductor pattern (including terminal areas) from the base laminate.

c. Blisters.

d. Measling or crazing.

e. Delamination of the base material.

f. Wrinkles in the conductor pattern.

g. Dirt, grease, or other foreign matter on the printed wiring boards.

h. Pits or inclusions.

4.18.11 Mounting of component parts. Components shall be mounted in accordance with DOD-STD-2000-2.

4.18.12 Insulation clearance. Clearance between the solder of the connection and the end of either separable or fixed insulation on the wire in the connection shall be as follows:

a. Minimum clearance. There shall be visible clearance between the insulation and solder connection. The insulation shall not abut the solder nor shall it be embedded in or surrounded by the solder. Neither shall the insulation be melted, charred, seared, nor diminished in diameter.

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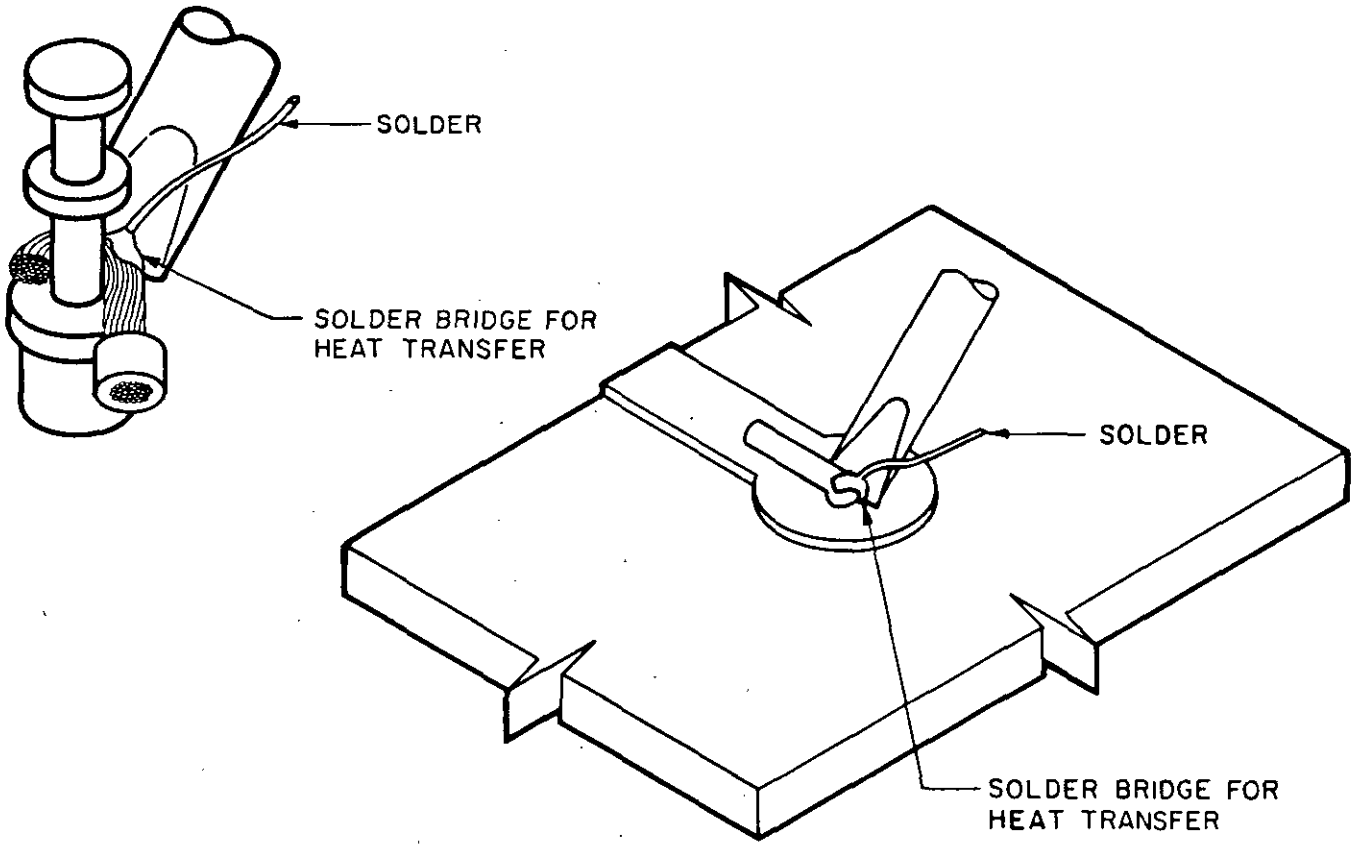
5.1.4 Flux application. Flux in accordance with 4.16.2 shall be utilized for all soldered electrical connections. Liquid flux, when used, shall be applied in a thin, even coat and shall be limited to those surfaces to be joined prior to application of heat. The use of excess flux shall be avoided. When used, cored solder shall be placed in such a position that allows the flux to flow and cover the connection elements as the solder melts. When an external liquid flux is used in conjunction with flux cored solders, the flux shall be of the same manufacturer's type.

5.1.5 Heat application. The elements to be soldered shall be sufficiently heated to cause melting of the solder and wetting of the surface. Excessive heating time, pressure, and temperature shall be avoided to prevent unreliable joints or damage to parts, printed circuitry, insulation, or adjacent components.

5.1.6 Solder application. The area to be soldered shall first be properly heated and then the solder, in accordance with 4.16.1, shall be applied at the junction of the soldering iron tip and the parts or components being soldered (see figure 7). When soldering printed wiring assemblies with plated-through holes, solder shall only be flowed from the solder side of the printed wiring assembly. As an exception, the use of transfer soldering techniques shall be approved by the Government Contracting Officer prior to use. Precautions shall be taken to prevent any change in wire or lead positioning within the solder connection before the solder has completely solidified. The solder bridge for heat transfer shall be maintained throughout the soldering process. Wicking of solder underneath the insulation of stranded wire during the soldering operation shall be minimized through use of anti-wicking tools, heat sinks, or other appropriate methods and techniques. As a guide the soldering operation shall be completed within 2 to 5 seconds.

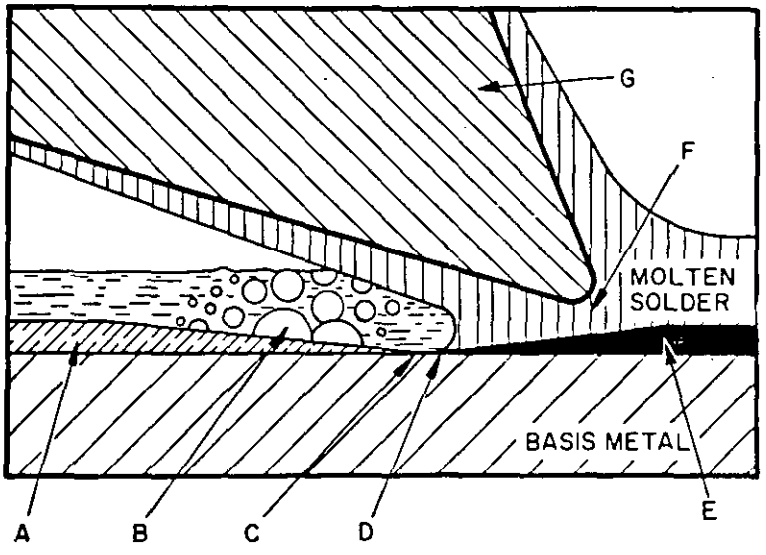
5.1.7 Cooling. The molten solder shall be cooled at room temperature only; forced air of liquid material shall not be used to accelerate solidification.

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INITIAL BRIDGING BEING FORMED

A



A. OXIDE FILM ON BASIS METAL.

B. BOILING FLUX SOLUTION REMOVING THE FILM OF OXIDE

C. BARE METAL IN CONTACT WITH FUSED FLUX.

D. LIQUID SOLDER REPLACING FUSED FLUX.

E. TIN REACTING WITH THE BASIS METAL TO FORM A NEW ALLOY.

F. SOLDER BRIDGE FOR HEAT TRANSFER

G. SOLDERING IRON TIP SHOWN NOT TOUCHING BASIS METAL FOR CLARITY.

B

FIGURE 7. Solder application and bridging (see 5.1.6).

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5.2.2.4 Lead trimming. Leads may be trimmed provided the cutters used impart no physical shock to the components and the leads are cut perpendicular to the axis of the lead within 10 degrees (actual measurement is not required except for referee purposes) with no spikes, peaks, or drag. When lead cutting is performed after flow soldering, one repass over the solder pot shall be made to cover ends and to reflow the solder terminations to prevent metallurgical disturbances caused by side loading. When the number of leads cut after soldering is small, manual touch-up may be used to reflow solder, including coverage of lead ends.

5.2.2.5 Preheating. The areas of the PWB to be soldered shall be preheated to 160 to 220°F (71° to 105°C) as measured on the component side at the surface of the laminate base material where there is no metal. All subsequent temperature measurements shall be made at the selected point for each board type. A preheat schedule shall be prepared for each type of board (as identified by PWB assembly part number designations) to be soldered based upon thickness and density and shall include the proper heater settings for preheating the boards to the required temperature. The selected temperature from board-to-board (of the same type) as measured on test specimens, shall be maintained within 10°F (5.5°C) but not to exceed the specified limit. Printed wiring assemblies shall be preheated to a temperature compatible with the flux, conveyor speed, solder temperature, and time of board contact with the solder bath.

5.2.2.6 Solder application. Solder shall be applied to the printed wiring assembly through contact with a solder wave compatible with the type of soldering machine being used. The time of contact between any point of the printed wiring assembly and the solder shall not exceed 5 seconds without prior approval by the Government Contracting Officer and shall be preselected, dependent upon preheating, thickness of the board, number of connections and conductors and the type of components. The temperature of the solder shall be measured at a point 1/16 to 1/8 inch (1.58 to 3.17 mm) (actual measurement not required except for referee purpose) below the crest(s) of the wave(s) in the path of the board and shall be the preselect temperature (T) ±10°F (±5.5°C) (see 5.2.2.1). The height of the wave and depth of immersion shall be maintained to assure effective contact and solder pressure sufficient for hole fill and solder spread to terminal areas on the component side of the printed wiring assembly.

5.2.3 Planar (Reflow) soldering.

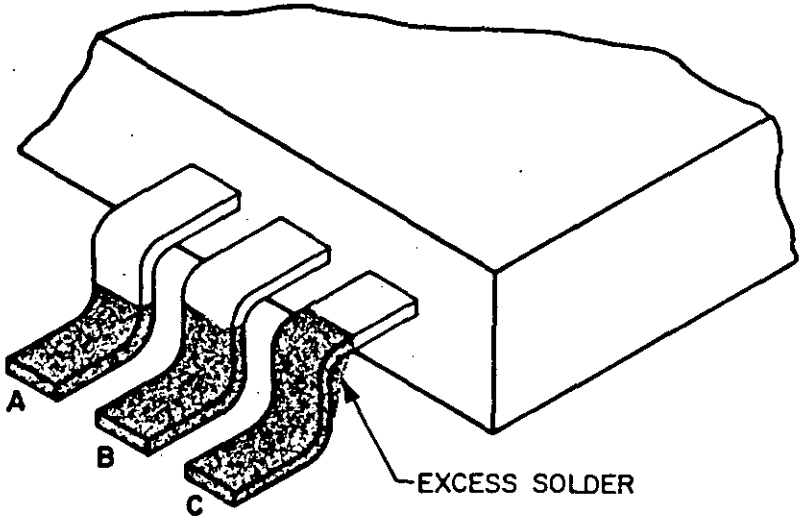
5.2.3.1 Solder application. Solder shall be deposited or plated on both metal surfaces to be joined prior to positioning the components in place.

5.2.3.2 Component positioning. Component mounting shall be in accordance with DOD-STD-2000-2.

5.2.3.3 Flux application. Prior to the soldering process, a uniform coating of flux, solder cream, or solder paste shall be applied to the pad surfaces or joint area where the component leads are to be soldered.

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5.2.3.4 Solder thickness. The thickness of the solder coating on component leads or ribbon conductors plus the solder thickness on the termination area shall be 0.0025 ± 0.0010 inches (0.064 ± 0.025 mm). The coating shall extend from the end of the ribbon lead up to and including the radius of the heel bend but shall not extend past the radius of the knee bend (see figure 11).



A - Minimum Acceptable: Extends past first radius
B - Maximum Acceptable: Tinning extends to second radius
C - Reject: Solder extends past second radius

FIGURE 11. Lead tinning of ribbon leads (see 5.2.3.4).

5.2.3.5 Heat application. The component terminations to be soldered shall be heated to the flow temperature of the solder. The application of heat shall be controlled during the soldering operation to prevent damage to the assembly (e.g., base material, adjacent connections, electrical components). The surfaces being soldered shall be restrained to preclude movement relative to the terminal area as solder is solidifying. Leads which are not seated in contact with the termination area shall not be deflected or forced into place during soldering such that residual stresses remain in the lead after soldering.

5.2.4 Condensation (Vapor Phase) soldering. The condensation (Vapor Phase) reflow soldering method is generally used for back planes and connector reflow soldering. Prior to using this method for component soldering, supportive information that no damage will occur to the components shall be submitted to the Government Contracting Officer. The condensation reflow system shall be mechanized to provide for smooth transition of the work piece, control of temperature, level of vapors and dwell time. There shall be a smooth transition out of the vapors after completion of the soldering operation to prevent disturbed solder connections prior to solidification.

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5.3 Post soldering operations.

5.3.1 Cleaning. Connections to terminals, chassis elements and printed wiring assemblies shall be cleaned using a solvent, cleaner or combination of solvents (see 4.16.4) within a time frame that affords total removal of contaminants and flux residue. The time between solidification of the solder and assembly cleaning shall not exceed 30 minutes except for connections that are hand soldered and interim cleaned during production assembly or during rework or repair operations. In those cases, removal of flux residue may be delayed for up to one hour after completion of soldering provided the final cleaning process used results in removal of all flux and contaminants from the finished product. Printed wiring assemblies, terminal board assemblies to which electrical components are added, and electrical/electronic modules shall be cleaned in a manner that will prevent both thermal shock to the assemblies, modules, or components thereon or therein and moisture intrusion into components not totally sealed. The cleaning shall be accomplished in accordance with 5.3.1.1, 5.3.1.2, 5.3.1.3, 5.3.1.4 or combinations thereof.

5.3.1.1 Dip tanks. Printed wiring assemblies, terminal board assemblies, and electrical or electronic modules shall be subjected to three separate cleaning baths each of which contain a solvent or blend of solvents (see 4.16.4) identical to that in the other two. The solvent in the tanks shall be changed in accordance with the schedule detailed below or the solids contents of the solvent in tank 3 shall be measured each 8-hour work shift and the solvents changed before the solids content reaches 10 ppm or when it is evident that the solids content will exceed 10 ppm before the next immediate measurement is scheduled. Immersion time in each tank shall be limited to 60 seconds. To change solvents, the solvent in tank 1 shall be discarded; the solvent in tank 2 may be placed in tank 1; the solvent in tank 3 may be placed in tank 2 and fresh solvent shall be added to tank 3.

a.

Order of cleaning	Change of solvent
Tank 1: first cleaning	After 2,500 square inches (1.6 square meter) of board surface per quart (0.95 liters) of solvent.
Tank 2: second cleaning	After 2,500 square inches (1.6 square meter) of board surface per quart (0.95 liters) of solvent.
Tank 3: third cleaning	After 2,500 square inches (1.6 square meter) of board surface per quart (0.95 liters) of solvent.

b. Use of a clean brush is permissible.

c. The printed wiring assembly shall be air dried after cleaning, using ambient air or compressed air which is oil-free and has a moisture content less than 2500 ppm measured at room ambient temperature. Air pressure up to 40 pounds per square inch gauge (psig) (276 kPa) may be used, provided it does not damage the assembly.

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5.3.1.2 Vapor degreasing. Use of vapor degreasing for cleaning printed wiring assemblies, terminal board assemblies, soldered chassis elements, and electrical or electronic modules is permissible provided that solvents are in accordance with 4.16.4 and are compatible with the particular vapor degreaser utilized for cleaning. The solid contents of the solvent in the (cold or final) sump of the degreaser shall not exceed 10 ppm.

5.3.1.3 Detergent cleaning. The use of detergent or saponifying cleaning systems is permissible for cleaning printed wiring assemblies and electrical or electronic modules only if the detergent or saponifier and the cleaning process are approved in advance by the Government Contracting Officer.

5.3.1.4 Ultrasonic cleaning. The use of ultrasonic cleaning devices on components or printed wiring assemblies is prohibited except for assemblies containing only terminals or connectors without internal electronics. Ultrasonic cleaning may be used on bare printed wiring boards.

5.3.2 Handling. After cleaning, printed wiring assemblies shall be handled in accordance with 5.3.7.

5.3.3 Cleanliness testing. After cleaning in accordance with 5.3.1, the printed wiring assemblies, terminal boards, and electrical/electronic modules shall be subjected to a cleanliness test before the end of the production shift. The resistivity of solvent extract test, or the sodium chloride (NaCl) salt equivalent ionic contamination test, or a test approved by the Government Contracting Officer shall be used to test for ionic cleanliness. The resistivity of solvent extract test must have a final value greater than 2,000,000 ohm-centimeters. The sodium chloride salt equivalent ionic contamination test must have a final value less than 10.0 micrograms per square inch of board surface area. One representative sample of each printed wiring assembly (PWA) type processed during each production shift, shall be subjected to the cleanliness verification test. Each type of printed wiring assembly shall consist of either the assemblies having the same part number or of different part numbered assemblies having the same size or assembly configuration using the same volume of resistivity solution. If the sample fails the cleanliness test, all lots represented by the failed sample shall be recleaned, retested and sampled, as a minimum, in accordance with MIL-STD-105, Inspection Level S-2, AQL-1.0 percent defective. Two subsequent lots shall be tested in accordance with MIL-STD-105 inspection level S-2, AQL 1.0 percent defective.

5.3.3.1 Resistivity of solvent extract. Solvent extract resistivity shall be measured as follows (see table IV):

a. Prepare a test solution of 75 percent by volume ACS reagent grade isopropyl alcohol and 25 percent by volume deionized water. Pass this solution through a mixed bed deionizer cartridge. After passage through the cartridge, the resistivity of the solution shall be greater than 6×10^6 ohm-cm (conductivity less than 0.166 microhmos/cm).

b. Clean a funnel, wash bottle, and a container with a portion of this test solution. Measure 10 milliliters of fresh test solution for each square inch of assembly area into the wash bottle. Assembly area includes the area of both sides of the board.

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c. Slowly, direct the test solution, in a fine stream, onto both sides of the assembly until all of the measured solution has been used.

5.3.3.2 Sodium chloride salt equivalent ionic contamination test. Sodium chloride salt equivalent ionic contamination shall be measured as follows (see table IV):

a. The sodium chloride salt equivalent ionic contamination test must use a solution of 75 percent isopropyl alcohol/25 percent deionized water. This solution must be verified for correct composition upon initial use and every four hours during a shift.

b. The equipment must be validated using a known amount of sodium chloride standard on the same schedule as the percentage composition verification.

c. The starting, or reference, purity of the solution must be greater than 20 x 10^6 ohm-centimeters (0.05 micromhos/centimeter) before each sample is tested.

TABLE IV. Cleanliness test values.

Test method	Starting resistivity	Ending value
Solvent Extract Resistivity	6 x 10^6 ohm-cm	shall be greater than 2 x 10^6 ohm-cm
Sodium Chloride Salt Equivalent Ionic Contamination	20 x 10^6 ohm-cm	shall be less than 10.0 micrograms/square inch

5.3.4 Initial inspection. After soldering and cleaning, each item shall be inspected in accordance with 5.4.4.

5.3.5 Rework of unsatisfactory solder connections. Rework operations shall be performed only by assembly personnel (operators) certified in accordance with 5.5.1. Inspection personnel shall perform no soldering or rework operations. Rework of unsatisfactory solder connections shall not be performed until inspection personnel have documented the discrepancy. This data shall be used to provide an indication as to possible causes and to determine the corrective action necessary to reduce future rework. When solder connections do not meet acceptance criteria specified herein, disposition of solder defects shall be as specified in 5.4.4.2.

5.3.5.1 Methods for rework by hand soldering.

5.3.5.1.1 Reheat method of rework. When rework is required, one reheating of a solder connection and adding flux and solder if necessary may be used for the following types of defects:

a. Nonsoldered connections.

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b. Cold solder connections.

c. Fractured or disturbed connections.

d. Insufficient solder.

e. Poor wetting.

f. Pits, holes, or voids in connection.

g. Visible basis metal in solder connection.

h. Solder points, peaks, or icicles.

5.3.5.1.2 Resolder method of rework. When rework is required, defective solder connections and other defects which cannot be corrected in accordance with 5.3.1.1.1 shall be reworked by removing the solder by a vacuum device or wicking. After the solder has been removed, the connection shall be cleaned and the defect corrected. All connections shall be cooled to room temperature before applying heat. Connections shall be resoldered in accordance with 5.1.6 and shall only be resoldered one time. If Level A defect(s) of Table V (see 5.4.4) still exist, process in accordance with 5.4.4.1.2.

CAUTION: Extreme care should be exercised when wicking solder out of plated-through hole.

5.3.5.2 Machine resolder method of rework. Assemblies may be resoldered by machine soldering a maximum of one time in accordance with 5.2. Assemblies that have been resoldered by machine shall not be reworked by machine soldering. Assemblies which are initially soldered by machine soldering in accordance with 5.2.2 or 5.2.3 and which contain defective solder connections, may be reworked by passing them through the same machine soldering process, with process control corrections, a single time to reflow all connections. The requirements of 5.2.2 or 5.2.3 shall be adhered to during the re-soldering. This method shall only be used when disposition of solder defects is in accordance with 5.4.4.2.3.

5.3.6 Post rework cleaning. After rework, printed wiring assemblies shall be cleaned in accordance with 5.3.1 and tested in accordance with 5.3.3.

5.3.7 Handling and storage after cleaning. After final cleaning, and prior to applicable of conformal coating or encapsulation, unfinished printed wiring assemblies shall not be handled such that areas to be conformally coated or encapsulated come in contact with bare hands or visibly contaminated tools. Such assemblies shall be stored and processed in a controlled environment (see paragraph 4.11.1.2).

5.3.8 Visual inspection of reworked defects. Each reworked defect including each resoldered or reheated connection shall be reinspected in accordance with 5.4.4 and shall conform to the acceptance criteria of DOD-STD-2000-3. Assemblies which have been reworked in accordance with 5.3.5.2 shall be completely reinspected for defects in accordance with 5.4.4 and DOD-STD-2000-3. The disposition of defects shall be in accordance with 5.4.4.2.3.

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5.3.9 Cleanliness testing prior to conformal coating or encapsulation. Prior to conformal coating or encapsulation, one representative sample of each printed wiring assembly (PWA) type processed during each production shift, shall be subjected to the cleanliness verification test. Each type of printed wiring assembly shall consist of either the assemblies having the same part number or of different part numbered assemblies having the same size or assembly configuration using the same volume of resistivity solution. If the sample fails the cleanliness test the entire lot shall be recleaned sampled in accordance with MIL-STD-105, inspection level S-2, AQL 1.0 percent defective and retested in accordance with 5.3.3 prior to conformal coating.

5.3.10 Conformal coating process. To facilitate the removal of defective components, the printed wiring assembly should be functionally tested prior to the conformal coating or any encapsulation process.

5.3.11 Conformal coating. Unless applicable Government-approved assembly drawings include a specific note stating that conformal coating is not to be applied, printed wiring assemblies shall be conformally coated with material conforming to MIL-I-46058 and in accordance with 5.3.11.1 through 5.3.11.8 below. The coating shall be applied to both sides of the printed wiring assemblies using either a brush, dip, spray or vacuum deposition system which is suitable to the particular assembly.

5.3.11.1 Conformal coating application. Conformal coating material shall be as specified on the Government approved assembly drawing. The material specification and manufacturer's instructions, as applicable, shall be followed and the material shall be used within the time period specified (both shelf-life and pot life). The coating shall be a minimum thickness (see 5.3.11.6) and fillets shall be kept to a minimum consistent with the size, weight, and mounting provisions of the components. A continuous coating shall be applied to both sides of the printed wiring assembly, using either a brush, dip, spray or vacuum deposition method which is suitable to the particular assembly and within the following criteria (see paragraphs 5.3.11.2 through 5.3.11.8). The coating shall be applied without bubbles, blisters, or breaks which might affect the printed wiring assembly operations, or sealing properties of the conformal coating.

5.3.11.2 Conformal coating on adjustable components. Assemblies having adjustable components shall not have the adjustable portion covered with the coating. Shafts of adjustable components shall be adjusted and sealed in accordance with 5.3.12.2.

5.3.11.3 Conformal coating on connectors. Mating connector surfaces of printed wiring assemblies shall not be coated with the conformal coating. Conformal coating shall, however, provide a seal between the connector and board and around all mounting devices of the connector.

5.3.11.4 Conformal coating on brackets. Printed wiring assemblies having brackets or other mounting devices shall not have the mating surfaces of said devices coated with conformal coating. However, the junction between these devices and the board and all attaching hardware shall be coated to provide a seal between the mounting device and the board.

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5.3.11.5 Conformal coating thickness. When thickness is not specified on the Government approved assembly drawing or material or process specification the coating thickness for conformal coating, as defined in MIL-I-46058 for Types AR, ER and UR shall be 0.002 ± 0.001 inch (0.05 ± 0.025 mm). For Type SR, the coating thickness shall be 0.005 ± 0.003 inch (0.127 ± 0.076 mm). When Type XY is required by the approved assembly drawing the coating thickness shall be 0.0005 to 0.0012 inch (.013 to .03 mm). All thicknesses are measured on a flat unencumbered surface.

5.3.11.6 Conformal coating on flexible leads. Components which are electrically connected to the printed wiring assembly by flexible leads shall have, as a minimum, the junction of the leads with the components and the printed wiring assembly coated.

5.3.11.7 Perimeter coating. Unless otherwise specified on the government approved assembly drawing, the outer perimeter of printed wiring assemblies shall not be increased in total thickness, by more than .040 inch (1.02 mm) as a result of conformal coating. The "outer perimeter" is defined as the area on each side of the board a distance of not more than 1/4 inch (6.35 mm) inward from the outer edge.

5.3.11.8 Edge coating. Unless otherwise specified on the government approved assembly drawing, the dimensions of the printed wiring assemblies shall not be increased in length or width by more than 0.030 inch (0.762 mm) on each edge, total 0.060 inch (1.524 mm), by application of conformal coating.

5.3.12 Adjustable components.

5.3.12.1 Adjustment. Adjustable components which incorporate integral adjustment locking mechanisms may be adjusted and locked prior to conformal coating. All other adjustable components shall be adjusted and staked after the assembly has been conformally coated.

5.3.12.2 Staking of adjustable components. After post-encapsulation adjustment, the external shafts of adjustable components which do not incorporate integral locking mechanisms shall be mechanically staked or sealed with pthalic alkyd resin enamel in accordance with MIL-E-22118. The sealant shall be used unthinned and only enough material to make the seal should be used.

5.3.13 Repair after conformal coating process. All processes which necessitate removal of conformal coating or encapsulation shall be classified as a repair operation.

5.4 Inspection. The contractor shall be responsible for conformance to the product and process requirements specified in the DOD-STD-2000 series of standards and shall also perform all process and product inspections and tests required by the DOD-STD-2000 series either directly or by subcontract. The contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless such other facilities are disapproved by the Government. The Government reserves the right to perform an audit or survey addressed to any requirements set forth in

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the DOD-STD-2000 series when such an audit is deemed necessary to assure that the supplies and services conform to the prescribed requirements. The contractor shall prepare and maintain process control procedures for all processes, inspections and tests required in the series of standards. These procedures shall provide sufficient detail to allow accomplishment by personnel of the appropriate skill level. They shall also delineate the accept/reject criteria and any special inspection requirements.

5.4.1 Surveillance. A surveillance program shall be conducted to determine that compliance with DOD-STD-2000-1, DOD-STD-2000-2 and DOD-STD-2000-3 series is being maintained, and to observe the control and disposition of nonconforming material. This includes periodic inspection of the work area, tools, materials, procedures and processes.

5.4.2 Inspection of tools and equipment. Tools and equipment shall be inspected for conformance to the requirements of 4.10, 4.11, 4.12, 4.13, 4.14 and 4.15. In addition, heat sources (see 4.11.2.9), soldering irons (see 4.11.3.1), solder pots (see 4.11.3.4), and process equipment (see 4.11.3) shall be tested in accordance with 5.4.2.1, 5.4.2.2, 5.4.2.3 and 5.4.2.4, respectively. Users should check hand tools daily for proper operation and cleanliness.

5.4.2.1 Heat sources tests. Heat sources shall be checked by the operator daily to ensure proper operation, heat range, and overall maintenance. The heat range may be checked by performance in lieu of taking actual temperature measurements. Nozzles, air deflectors and adapters, input power cords, and plugs on cords shall be examined and there shall be no condition that could cause damage to items to be processed (see 4.11.3).

5.4.2.2 Soldering irons. Soldering irons shall be tested subsequent to receipt after purchase and prior to issue for soldering operations and shall meet physical and functional characteristics in accordance with 4.11.3 and 4.11.3.1. Soldering irons in use shall be tested at periodic intervals as defined in 5.4.2.2.2 and 5.4.2.2.3. When not retained in stores, each soldering iron shall be examined in accordance with 5.4.2.2.1 prior to use during any day the iron is in use. If repaired or modified, soldering irons shall be retested and shall conform to all requirements of 4.11.3 and 4.11.3.1 before the iron is utilized for production use.

5.4.2.2.1 Daily check by operator. The following shall be performed daily by the operator. Before the iron is energized, the handles, input power cord, power cord plug, and other elements shall be examined for physical damage and there shall be no conditions that could result in defective connections and in damage to components and other items to be soldered. Tips and heating elements shall be examined and there shall be no oxidation, corrosion, or foreign materials that restricts or impedes mating between tip and heating element or heating element and iron. Irons and tips shall be examined for compatibility with the soldering operations to be performed and size and shape shall be as specified in 4.11.3.1. Operators shall ensure the proper operation and performance of heated irons (see 5.1.6).

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5.4.2.2.2 Quarter-year tests. The following tests shall be performed quarterly. The intervals shall be shortened as required to assure continued conformance with the specified requirements as evidenced by the results of preceding periodic measurement tests and may be lengthened only when the results of such prior periodic tests provide definite indication that such action will not result in use of out-of-conformance irons. Tip-to-ground resistance shall be measured and shall conform to 4.11.3 and 4.11.3.1. The tip temperature of non-adjustable temperature-controlled irons shall be measured and shall be within ±10°F (±5.5°C) of the rated (idling) temperature (see 4.11.3.1). The tip temperature of adjustable temperature-controlled irons shall be measured, as a minimum, at three separate pre-selected tip temperatures with one each at a low, middle, and high temperature in the adjustable range of the iron. The tip temperature shall be within ±10°F (±5.5°C) at each of the three preselected temperatures.

5.4.2.2.3 Semi-annual tests. The following tests shall be performed semi-annually. The intervals shall be shortened as required to assure continued conformance with the specified requirements as evidenced by the results of preceding periodic measurement tests and may be lengthened only when the results of such prior periodic tests provide definite indication that such action will not result in use of out-of-conformance irons. Soldering irons shall be tested for potential differences between tip and ground and the difference shall be no greater than as specified in 4.11.3.1 when measured as specified in 4.11.3.

5.4.2.3 Solder pot tests. Stand-alone solder pots and solder pots integral to machine soldering systems shall be tested subsequent to purchase and receipt of the pot or machine and prior to initial use in soldering operations (including tinning and solderability testing). The pots shall conform to 4.11.3 and 4.11.3.4 when tested. Stand-alone solder pots and those integral to machine soldering systems shall be tested at 6-month intervals for pot solder-to-ground resistance (see 4.11.3), for solder temperature (see 4.11.3.4), and for electrical potential difference between pot solder and ground (see 4.11.3). The 6-month intervals shall be shortened as required to assure continued conformance with specified requirements as evidenced by the results of preceding periodic tests and may be lengthened only when the results of such previous periodic tests provide a definite indication that such action will not result in use of out-of-conformance pots. Solder pots shall also be examined daily for overall maintenance. The input power cord, the input power plug, connections to terminal lugs (if any), and other physical attributes of the pot shall be examined and there shall be no condition that could result in damage to work in process.

5.4.2.4 Process equipment. Process equipment shall be tested subsequent to purchase and receipt and prior to initial production use and shall conform to requirements of 4.11.3. Except for tests of integral solder pots (see 5.4.2.3), process equipment shall be tested annually to assure that resistance between any item to be processed and ground is no greater than specified, that electrostatic discharge protection is as specified in 4.11.3, and that potential difference between items to be processed and ground is no greater than specified in 4.11.3. The interval between tests shall be shortened from annual to a shorter span as required to assure continued conformance with specified requirements as evidenced by the results of preceding periodic tests

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and may be lengthened only when the results of such preceding periodic tests provide a definite indication that such lengthening will not result in use of out-of-conformance process equipment.

5.4.3 Inspection of material. All materials listed in 4.16 shall be inspected for compliance with the requirements of 4.16.1 through 4.16.10. Material controls shall be implemented to ensure that only conforming materials, tools and equipment are used. Materials, tools and equipment not conforming or not required for the operations involved shall be removed from the work area.

5.4.3.1 Inspection of printed wiring boards. Printed wiring board shall be inspected and tested for conformance to 4.16.8 or 4.16.9 as modified by 4.16.10. Sampling inspection shall be as delineated in MIL-P-55110 or MIL-P-50884, respectively, except that the test to verify the copper plating process specified in 4.16.8.1 and 4.16.9.1 shall be performed as a minimum every two weeks using test methods as follows:

a. Copper purity in accordance with ASTM E 53.

b. Tensile strength (vertical and horizontal) in accordance with ASTM E 8 or ASTM E 345.

c. Elongation (vertical and horizontal) in accordance with ASTM E 8 or ASTM E 345.

5.4.3.2 Inspection records. Accurate records shall be kept which provide evidence that the materials and tools are either listed on a Qualified Parts List (QPL) or that sufficient testing has been performed to satisfy the requirements of the applicable material specification and any additional requirements specified herein.

5.4.4 Inspection of soldered connections and assemblies. Each assembly which includes soldered connections and each soldered connection shall be one hundred percent visually inspected (see 4.4) to assess conformance with the requirements herein, the requirements of DOD-STD-2000-2 and DOD-STD-2000-3, and the requirements of the applicable assembly drawings.

5.4.4.1 Defects and defect rates. Each variance from requirements of the DOD-STD-2000-1, DOD-STD-2000-2 and DOD-STD-2000-3 series or from requirements of applicable drawings is a defect which shall be documented whether the defect is one that can be returned to full conformance (reworkable defect), one that cannot be returned to full conformance but can be returned to usable and acceptable condition through more extensive action than rework (repairable defect), or one that cannot be returned to usable and acceptable condition either by rework or repair. All defects including those listed in table V shall be documented prior to any rework or repair action (see 5.3.5) and prior to any submission for Material Review Board (MRB) decision (see 5.4.4.1.3). An assembly defect rate and a solder defect rate shall be calculated in accordance with 5.4.4.1.4 for all printed wiring assemblies and terminal boards. However, the defect rate calculation for any other type of assembly may be optional. If the option to forego defect rate calculation is exercised for the other type, the option shall be continued for the

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contract of such assemblies. When defect rates are not calculated, all defects shall be treated as Level A of table V. All defects not listed in table V shall be designated Level A defects.

5.4.4.1.1 Reworkable defects. After inspection and documentation, reworkable defects shall be returned to the proper area for rework, testing if required, and reinspection (see 5.4.4.2.2c and 5.4.4.2.3c).

5.4.4.1.2 Nonreworkable defects. Nonreworkable defects shall be submitted to a proper constituted Material Review Board (MRB) when authorized by contract. As a minimum, one contractor member of the MRB shall be certified to Category C requirements (see 5.5.3.3) or Category H requirements (see 5.5.3.8) and one Government member, when required, shall be certified to Category B requirements (see 5.5.3.2) or Category H requirements.

5.4.4.1.3 MRB decisions. MRB decisions are limited to scrap, repair and use as is. Use as is disposition requires Government concurrence and the rationale for acceptance shall be completely documented. Repair disposition shall be as follows:

a. Repair in accordance with a previously documented and Government approved repair procedure.

b. A Government approved special repair not previously documented. This repair procedure shall be documented step-by-step and shall include acceptance and rejection criteria.

After the maximum of one reheat has been accomplished and Level A defects remain, further rework shall not be authorized. This condition shall be categorized as nonreworkable and may only be subsequently returned to compliance by repair disposition and utilizing a Government approved repair procedure. This disposition requires Government concurrence, rationale for use, and shall be completely documented. The repaired area shall be reinspected and shall conform to the requirements herein, DOD-STD-2000-2, DOD-STD-2000-3, and the assembly drawing.

5.4.4.1.4 Defect rate calculation. The defect rate (see 5.4.4.1) is the number of observed defects divided by the normalizing number. The assembly defect rate and solder defect rate shall be calculated and documented each time inspection, in accordance with 5.4.4 or 5.3.8, is performed. Assembly defect rates and solder defect rates are separately calculated as follows:

a. The assembly defect rate shall be calculated by dividing the assembly defects observed (including table V defects) by the normalizing number. The normalizing number for assembly defects shall be the total of the number of components, component leads, terminals and wires contained on the assembly plus one for the printed wiring board.

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b. The solder defect rate shall be calculated by dividing the solder defects observed (including table V defects) by the normalizing number. The normalizing number for solder defects shall be the total number of terminals plus the total number of solder connections (for plated-through-hole printed wiring boards the top and bottom connection is ONE integral connection) on the assembly. For calculating hand solder defect rates, only hand soldered connections shall be counted. For calculating machine solder defect rates, only machine soldered connections shall be counted. No connection shall be counted twice.

c. The assembly and solder defect rates shall be calculated and documented for each assembly. In addition, the summation of each defect rate shall be calculated and documented each working day for all assemblies inspected that day. Defect rate documentation and data shall be made available to the Government.

NOTE: For ease of understanding defect calculations and dispositions, a decision chart and sample calculations are provided as Appendix C.

5.4.4.2 Disposition of defects. The disposition of defects, including but not limited to those defects listed in table V, shall be as follows:

a. For hand soldered assemblies, the disposition of defects shall be in accordance with 5.4.4.2.1 and 5.4.4.2.2.

b. For machine soldered assemblies, the disposition of defects shall be in accordance with 5.4.4.2.1 and 5.4.4.2.3.

5.4.4.2.1 Disposition of assembly defects (nonsolder defects). This procedure shall apply to all assemblies, both hand soldered and machine soldered. The disposition of assembly defects shall be as follows:

a. The disposition of Level A defects A180 thru A195 and A198 of table V shall be in accordance with 5.4.4.1.2.

b. Level A defects A130 thru A160 and A196 and A197 of table V shall be rejected and reworked in accordance with 5.3.5.1 or processed in accordance with 5.4.4.1.2.

c. Level B defects need not be reworked except that when the assembly defect rate exceeds 3.0 percent, Level B defects shall be reworked in accordance with 5.3.5.1 and process corrective action shall be taken. When the defect rate is less than 3 percent, only Level A defects shall be reworked.

5.4.4.2.2 Disposition of hand soldering defects. This procedure shall apply to all hand soldering. The disposition of defects shall be as follows:

a. Defects as listed in Level A of table V shall be rejected and reworked in accordance with 5.3.5.1 or rejected and processed in accordance with 5.4.4.1.2.

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b. Level B defects need not be reworked except when the solder defect rate exceeds 3.0 percent the defects shall be reworked in accordance with 5.3.5.1 and process corrective action shall be taken to preclude recurrence of the defects during further production. When the defect rate is less than 3 percent, only Level A defects shall be reworked.

c. After having been reworked a single time in accordance with the procedure of 5.3.5.1, assemblies which contain defects shall be rejected and processed in accordance with 5.4.4.1.2 if there still is:

(1) a Level A defect as listed, in table V; or

(2) an assembly defect rate greater than 3.0 percent; or

(3) a soldering defect rate greater than 3.0 percent.

5.4.4.2.3 Disposition of machine soldering defects. This procedure shall apply to all machine soldered assemblies. The disposition of defects shall be as follows:

a. Machine soldered assemblies which have a solder defect rate of 3.0 percent or less and contain defects listed as Level A of table V, or which contain 200 or less solder connections with a maximum of 10 defects, shall be reworked using one of the following methods.

Method 1. Rework the Level A defects only, in accordance with 5.3.5.1.

Method 2. Rework in accordance with 5.3.5.2.

Method 3. Process in accordance with 5.4.4.1.2.

b. Machine soldered assemblies which have a solder defect rate greater than 3.0 percent shall be reworked in accordance with 5.3.5.2 or processed in accordance with 5.4.4.1.2. Corrective process control action shall be taken to preclude recurrence of the defects during further production.

c. After having been reworked a single time in accordance with the procedure of 5.3.5.2, assemblies which contain defects shall be rejected and processed in accordance with 5.4.4.1.2 if there still is:

(1) a Level A defect as listed in table V; or

(2) an assembly defect rate greater than 3.0 percent; or

(3) a solder defect rate greater than 3.0 percent.

d. Assembly defects shall be corrected in accordance with the requirements of 5.4.4.2.1 before resoldering in accordance with 5.3.5.2.

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TABLE V. Defects.

No.	Level A Soldering Defects
A101	Nonsoldered connections
A102	Bridging
A103	Rosin connection
A104	Cold solder joint
A105	Fractured or disturbed
A106	Insufficient solder
A107	Excessive solder, lead not discernible
A108	Poor wetting
A109	Solder splattering
A110	Pits, pinholes, holes, or voids
A111	Dewetting on PWB conductor paths (areas which are not part of a solder joint)
A112	Dewetting of solder connection area(s)
A113	Solder not smooth and shiny
A114	Solder points, peaks, or icicles
A115	Flux residue, oils, greases on assembly
A116	Improper tinning of stranded wire
A117	Overheated solder
A118	Unfilled plated-through hole
A119	Contaminants in solder connection
A120	Solder in bend radius on both sides of components
A121	All other solder defects except those listed as Level B defects
No.	**Level A Assembly Defects**
A130	Improper transmission of stress of leads and wires (improper stress relief)
A131	Metal-cased components mounted over circuit path (not insulated)
A132	Uninsulated wires or component leads routed over the circuit path where shorting is possible
A133	Charring, burning, or damage to insulation (affecting electrical operation)
A134	Birdcaged wire
A135	Cut or nicked leads or wires
A136	Leads or wires scraped, exposing metal or stretched
A137	Lead clinched beyond allowable limits (electrical clearance insufficient)
A138	Excessive lead length (electrical clearance insufficient)
A139	Insufficient lead length
A140	Excessive wicking
A141	Improper insulation clearance
A142	Excessive terminal fill
A143	Improper lead bend radius
A144	Improper lead bend clearance

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TABLE V. Defects (Continued).

No.	Level A Assembly Defects
A145	Unclinched components where clinching is required
A146	Component has improper clearance to adjacent components
A147	Component has improper clearance above the PWB
A148	Component not sufficiently supported
A149	Glass-cased components not protected by buffer material when epoxy conformal coating is used
A150	Component less than 1/16 inch (1.6 mm) to edge of PWB causing interference
A151	Component obscures termination or another component (unless design dictates otherwise)
A152	Physical damage of components
A153	Improper vertical mounted component clearance
A154	Improper terminal swaging
A155	Leads cut after final soldering
A156	Wrong parts used
A157	Wrong orientation of polarized components
A158	Component improperly spaced or located
A159	Damage to wire or strands
A160	All other assembly defects except those listed as Level B defects
No.	**Level A Soldering, Assembly and Printed Wiring Boards Defects**
	Soldering and Assembly Defects of Printed Wiring and Printed Wiring Boards
A180	Pattern delaminated (after soldering)
A181	Burned, scorched PWB or parts
A182	Excessive warp or twist (after soldering)
A183	Measling or crazing (after soldering)
A184	Delamination of base material
A185	Separation of conductor pattern
A186	All other printed wiring and printed wiring boards defects except those listed as Level B defects
	Printed Wiring and Printed Wiring Boards Defects (prior to soldering)
A190	Excessive warp or twist
A191	Pits, scratches, or inclusions
A192	Blisters in conductor pattern
A193	Wrinkles in conductor pattern
A194	Improper hole size
A195	Measling or crazing
A196	Improper solder coating
A197	Slivers on circuit
A198	All other printed wiring and printed wiring boards defects except those listed as Level B defects

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TABLE V. Defects (Continued).

No.	Level B Defects
B201	Charring, burning or damage to insulation (not affecting electrical operation)
B202	Stranded wire lay disturbed (with no strand separation)
B203	Leads or wires scraped not exposing basis metal
B204	Insufficient or no lead clinch (non plated-through holes)
B205	Excessive lead length (electrical clearance sufficient)
B206	Improper lead wrap on terminals
B207	Component not resting on feet, pads or projections
B208	Component markings not legible with correct component installed
B209	Component not mounted perpendicularly
B210	Component less than 1/16 inch (1.6 mm) to edge of PWB not causing interference with mounting of PWA
B211	Possible component abrasion with another component
B212	Improper lead swaging
B213	Lack of solder coverage on lead ends (not due to cutting after solder or handling)
B214	Excessive solder, lead is discernible
B215	Component not centered
B216	PWB scratches (that do not expose basis metal or fiber)

5.5 Certification of personnel.

5.5.1 Certification. Certification as to the ability of personnel to meet the requirements of this standard, DOD-STD-2000-2 and DOD-STD-2000-3 shall be made available to the Government Contracting Officer. Personnel shall be certified in accordance with the requirements specified herein, prior to performing or inspecting soldering on engineering or experimental models, prototype models, or deliverable articles. Each individual's certification card shall be on his person or visibly displayed at his work station.

5.5.2 Visual acuity. All candidates for certification in any category (see 5.5.3) shall, as a minimum, meet the following vision requirements:

a. Far vision: Snellen chart 20/40, or better.

b. Near vision: Jaeger 1 of 0.50 mm letters at 14 inches (35.56 cm) or better.

c. Color perception: Normal as determined by means of standard color plates (i.e., Dvorine pseudo-isochromatic plates, Ishihara plates, or equivalent).

d. Depth perception: Normal binocular vision for fine, close work as determined by standard testing methods.

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All personnel certified to this standard shall be given an eye test by an accredited eye examiner to determine their ability to meet these vision requirements. Frequency of eye tests shall be annually for persons 35 years of age or older and every two years for persons less than 35 years of age, except frequency for an individual may be increased if so prescribed by a licensed ophthalmologist. Use of prescription lenses to meet the vision requirements is permissible. When such lenses are required, the certification card shall so state and such lenses shall be used whenever soldering or inspection is being performed. When a binocular microscope, with individually adjustable eyepieces, is used for the soldering or inspection operation, prescription lenses are not required. Failure to meet any of these requirements shall disqualify a candidate for training and certification or shall result in revocation of any certification previously granted.

5.5.3 Certificate categories. Certificates shall be issued in the following ten categories:

A	Government Senior Examiner
B	Government Instructor and Examiner (except category M)
C	Contractor Instructor and Examiner
D	Inspector (Contractor)
E	Operator (Contractor)
F	Government Inspector (except category N)
G	Government Operator
H	Process Examiner
M	Government Contract Administration Office (GCAO) Instructor and Examiner
N	Government Contract Administration Office (GCAO) Quality Assurance Representative (QAR)
R	Restricted Operator or Inspector

Contractor personnel (category C) shall be certified by an approved school; contractor personnel (categories D, E, and R) shall be certified by the contractor.

5.5.3.1 Category A Government Senior examiner.

5.5.3.1.1 Category A Authority. Category A personnel are authorized to train, certify and require certification and recertification of personnel of all other categories. Category A personnel are also authorized to monitor and evaluate soldering processes, workmanship, training programs, and facilities for conformance to this standard.

5.5.3.2 Category B Government instructor and examiner (except category M).

5.5.3.2.1 Category B Authority. Category B personnel are authorized to require recertification of personnel of categories C, D, E, R, F, and G, to monitor contractor soldering processes and workmanship for conformance to this standard and to perform soldering operations and inspections at Government facilities for conformance with this standard.

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5.5.3.3 Category C Contractor instructor and examiner.

5.5.3.3.1 Category C Contractor personnel. Category C personnel shall be contractor personnel certified after satisfactory completion of an 80 hour soldering course (see 5.5.7.4). Category C personnel certification and performance shall be reviewed on an annual basis.

5.5.3.3.2 Category C Authority. Category C personnel are authorized to train or require recertification of personnel of categories D, E, and R provided that the personnel are in the employ of:

a. The same company, or

b. a division of the same company, or

c. a subcontractor of the same company as that which employs the category C person on a full time basis.

Category C personnel are authorized to monitor soldering processes and workmanship for compliance to this standard, to perform inspections for conformance with this standard, and to determine the operations or procedures that are appropriate for a category R inspector or operator.

5.5.3.4 Category D Inspector.

5.5.3.4.1 Category D Contractor personnel. Category D personnel shall:

a. Have completed all requirements for Category E operator as specified in 5.5.3.5, and

b. be certified by the contractor after satisfactory completion of an additional 20 hour minimum inspector course conducted by a certified Category C instructor and examiner.

5.5.3.4.2 Category D Authority. Category D personnel are authorized to perform inspections for conformance with this standard provided that the personnel are in the employ of:

a. The same company, or

b. a division of the same company, or

c. a subcontractor of the same company as that which employs the category C person on a full time basis.

5.5.3.5 Category E Operator.

5.5.3.5.1 Category E Contractor personnel. Category E personnel shall be certified by the contractor after satisfactory completion of a 40-hour soldering course conducted by a certified category C instructor and examiner.

5.5.3.5.2 Category E Authority. Category E personnel are authorized to perform soldering operations in conformance with this standard provided that the personnel are in the employ of:

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a. The same company, or

b. a division of the same company, or

c. a subcontractor of the same company as that which employs the category C person on a full time basis.

5.5.3.6 Category F Government inspector (except category N).

5.5.3.6.1 Category F Authority. Category F personnel are authorized to perform inspections for conformance with this standard.

5.5.3.7 Category G Government operator.

5.5.3.7.1 Category G Authority. Category G personnel are authorized to perform operations in conformance with this standard.

5.5.3.8 Category H process examiner.

5.5.3.8.1 Category H personnel. Category H personnel shall be certified by the Government after satisfactory completion of an 80-hour soldering course, with primary emphasis on inspection and machine processes. Category H personnel certification and performance shall be reviewed on an annual basis. Contractors are not required to have Category H personnel as all requirements for Category H personnel may be accomplished by Category C personnel.

5.5.3.8.2 Category H authority. Category H Government personnel are authorized to recommend recertification of category H contractor personnel and personnel of categories C, D, E and R, to inspect soldering processes and workmanship for conformance to this standard, to perform quality assurance actions, and to participate in materials review board (MRB) decisions. Category H contractor personnel are authorized to recommend recertification of contractor personnel of categories D, E and R, to inspect contractor soldering processes and workmanship for conformance to this standard, to perform quality assurance actions, and to participate in material review board decisions.

5.5.3.9 Category M GCAO instructor and examiner.

5.5.3.9.1 Category M Authority. Category M personnel are authorized to require recertification of personnel of categories C, D, E, R, and N, and to monitor contractor soldering processes and workmanship for conformance to this standard.

5.5.3.9.2 Category M Training authority. Category M personnel are authorized to train category N personnel to this standard.

5.5.3.10 Category N GCAO Quality Assurance Representative (QAR).

5.5.3.10.1 Category N Authority. Category N personnel are authorized to monitor and verify the certification status of Category C, D, E and R contractor personnel and to recommend recertification when appropriate. Category N personnel are authorized to monitor contractor soldering processes and workmanship to verify contractor conformance to these standards.

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5.5.3.11 Category R Restricted operator or inspector.

5.5.3.11.1 Category R Contractor personnel. Category R personnel shall be selected by the contractor for performance of a limited number of operations or procedures. The training program shall, as a minimum, include the operations or procedures for which the category R personnel are to be certified. Functions or operations using a category R operator or inspector at the contractor facility require prior approval of a Category A Government Senior Examiner. Similar operations conducted by a subcontractor requires approval of the prime contractor and is subject to review by category A or B personnel in accordance with 5.5.3.1.1 and 5.5.3.2.1.

5.5.4 Achievement of certified status. To be certified in category C, personnel shall attend and successfully complete an 80-hour formal training program at a Government approved school, as specified by the procuring activity. To be certified in categories D and E, personnel shall attend and successfully complete a 40-hour formal training program conducted by a category C instructor and examiner.

5.5.5 Training programs.

5.5.5.1 Contractor training program.

5.5.5.1.1 General. The contractor shall establish and maintain an effective written training program to qualify, certify and recertify all personnel performing operations applicable to this standard under the cognizant Government activity contracts and shall include subcontracts, interplant work orders, and purchase orders. The program shall provide for training using the methods, equipment and materials described in this standard. Upon satisfactory demonstration of proficiency, personnel shall be issued a contractor certificate. The contractor shall prepare and maintain records of personnel training and performance.

5.5.5.1.2 Training records. The contractor's training records shall be maintained for the term of the contract and shall include, for each trainee during the time of his or her certification, the following:

a. Trainee fabricated test specimen depicting satisfactory conformance to the applicable requirements of this standard.

b. Graded copies of written tests.

c. Employer, plant division, and location of employment.

d. Certification category.

e. Date of certification or recertification.

f. Records of latest visual acuity examination.

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5.5.5.1.3 Program evaluation. The training program is subject to review by a category A senior examiner to assure that the program is compatible with this standard. The category A senior examiner reserves the right to disapprove the training program at any time the objectives of this standard are not being met.

5.5.5.2 Government training program. The Government is responsible for certification of its personnel.

5.5.6 Maintenance of certified status.

5.5.6.1 General. Based on a quality audit, review of inspection data, or observation of quality of articles fabricated, soldering personnel involved may be required to either demonstrate proficiency or be retrained in the category concerned and be recertified (see 5.5.7).

5.5.6.2 Continuous performance evaluation. Category D and E personnel shall be subjected to a continuous performance evaluation as specified in 5.5.7.2.

5.5.7 Recertification.

5.5.7.1 Requirements. Recertification shall be required under the following circumstances:

a. Proficiency requirements herein are not met.

b. New techniques have been developed which require new skills.

c. Certificate holder changes employer.

d. There is a reason to question proficiency or workmanship.

e. Work period interruptions exceeding 90 days (categories C, D, E and R).

f. Twelve months after last certification.

5.5.7.2 Procedures. The recertification procedure shall be as follows:

a. Category C personnel shall annually attend a minimum 8-hour update course conducted by one of the certification centers (see 5.5.7.4) and shall demonstrate proficiency to a Category A senior examiner or designated Category B representative. Category C personnel failing to demonstrate proficiency shall be required to satisfactorily complete a certification course prior to further performance. Personnel passing the update course will have their certification extended for a period of one year. Every fourth year after initial certification, Category C personnel shall attend a minimum 40 hour recertification course held at one of the centers (see 5.5.7.4). The

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recertification course shall be in lieu of the annual update course. Upon satisfactory completion of the 40 hour course, Category C personnel will be recertified and continue in the update and recertification cycle.

b. When 5.5.7.1b applies, category C personnel shall be recertified after attending a recertification course, and demonstrating proficiency in new techniques.

c. Category D, E and R personnel shall be recertified as specified by category A, B, or C personnel in accordance with the requirements of 5.5.5.1.1. All recertification programs shall include a requirement to score a minimum of 80 percent correct on a written test compatible with that taken in the original certification course.

5.5.7.3 Revocation of certified status. Certifications issued by the contractor shall be revoked when the:

a. Certificate holder fails to be recertified when required.

b. Contractor training program fails to meet the objectives of this standard.

c. Certificate holder leaves employment of the contractor for which certificate was originally issued.

d. Certificate holder fails to meet visual acuity requirements.

e. Quality of work has deteriorated as determined by a category A, B, C, or M instructor and examiner.

5.5.7.4 Certification resources. Addresses and telephone numbers of approved schools may be obtained from:

Commander
Air Force Wright Aeronautical Laboratory
(AFWAL/MLSE)
Wright-Patterson AFB, OH 45433-6503
Telephone: Commercial (513) 255-5128
AUTOVON 785-5128

Commander
U. S. Army Missile Command
(AMSMI-SE-TD-ST)
Redstone Arsenal, AL 35898-5290
Telephone: Commercial (205) 876-9086
AUTOVON 746-9086

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Commanding Officer
Naval Air Engineering Center
(Code 9313)
Lakehurst, NJ 08733-5100
Telephone: Commercial (201) 323-7475
AUTOVON 624-7475

6. NOTES

6.1 Supersession note. This standard is one of a series of documents for High Quality/High Reliability soldering. The series consists of DOD-STD-2000-1 Soldering Technology, High Quality/High Reliability; DOD-STD-2000-2 Part and Component Mounting for High Quality/High Reliability Soldered Electrical and Electronic Assemblies; DOD-STD-2000-3 Criteria for High Quality/High Reliability Soldering Technology. As a set this series is intended to supersede the documents listed below. The listed documents will remain in effect until they are cancelled by separate notice.

MIL-S-45743	Soldering, Manual Type, High Reliability, Electrical and Electronic Equipment
MIL-S-46844	Solder Bath Soldering of Printed Wiring Assemblies
MIL-S-46860	Soldering of Metallic Ribbon Lead Materials to Solder Coated Conductors, Process for Reflow
MIL-S-50826(AR)	Soft Soldered Electrical Connections for Special Weapons Items Including Other Related Electronic Devices
MIL-S-50827(AR)	Soft Soldered Electrical Connections for Conventional Weapons Items Including Other Related Electronic Devices
MIL-STD-252(CR)	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and Other Devices
MIL-STD-1460(AR)	Soldering of Electrical Connections and Printed Wiring Assemblies, Procedures for
QWS-10.00B	Acceptance Criteria for Solder Connections and Wiring in Electronic Equipment
WS-4554	Wave Soldering of Printed Wiring Assemblies, Automatic Machine Type
WS-6536	Process Specification Procedures and Requirements for Preparation and Soldering of Electrical Connections
WS-14146	Preparation, Soldering and Inspection of Electrical Connections

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6.2 Changes from previous issue. The margins of this standard are marked with vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.3 Subject term (key word) listing.

Connector
Degreaser
Electronic components assembly
Flux, soldering
Insulation sleeving, electrical
Printed circuit board
Printed wiring, flexible
Printed wiring board
Solder
Soldering iron
Solvent, stoddard's
Wiring harness

6.4 Use of metric units. English units (inches, pounds, Fahrenheit) are the primary units used by the United States Industry for manufacturing electronic assemblies. In this document, measurements are provided in English units followed by the metric equivalent. The conversions from English to metric are made in accordance with FED-STD-376. The metric equivalents provided in this document are rounded to sensible values. Direct conversions which are mathematically correct but reflect unreasonable degrees of precision in metric units have been avoided (i.e., 5.002 mm is considered unreasonable; this should be rounded to 5.0 mm). In the event of conflict, the primary English measurement shall take precedence.

Custodian:
Air Force - 20
Army - MI
Navy - AS
NSA - NS

Review activities:
AF-11, 15, 17, 19, 84, 99
Army - AR, CR, ER, MR
Navy - EC, OS
DLA - ES, DH

User activities:
Army - AV
Navy - MC
DLA - DH

Preparing Activity:
Navy - AS
(Project No. SOLD-0022)

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APPENDIX A

PROCEDURE FOR MEASURING 2mV POTENTIAL AT SOLDERING IRON TIP
(See 4.11.3)

10. SCOPE. This appendix establishes the procedure for measuring 2 mv potential at the soldering iron tip.

20. REFERENCED DOCUMENTS. Not applicable.

30. DEFINITIONS. Not applicable.

40. GENERAL REQUIREMENTS.

40.1 Equipment requirements. Minimum use specifications are the principal parameters required to perform the measurement and are included to assist in the selection of alternative equipment which may be used at the discretion of the using laboratory. Voltmeters must bear evidence of end-date calibration.

TABLE A-I. Equipment requirements.

Item	Minimum use specifications	Measurement equipment and devices (see 40.2)
TRMS voltmeter	Accuracy: ± 10% at 2 mV RMS Frequency range: 50-500 Hz Input impedance: 10 megohms	Hewlett-Packard model 3400A or Keithley model 132F
Cable	Type: shielded BNC (male) to alligator	Pomona Electronics model AL-B-BNC-36
Ground clip	Type: oscilloscope probe to alligator	Tektronix part numbers 175-0125-01 and 344-0046-00
Shim stock	Material: brass or copper Thickness: 0.008-0.020 inches (0.20-0.5 mm) Size: approximately 1.5 x 0.75 (4.81 x 1.90 cm) inches	
Solder	Type: rosin core	

40.2 The instruments used in this procedure and listed in table A-I were selected from those known to be available at Department of Defense facilities, and the listing by make or model number carries no implication of preference, recommendation or approval by the Department of Defense for other agencies. It is recognized that equivalent equipment produced by other manufacturers may be capable of equally satisfactory performance in the procedure and may be used.

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APPENDIX A

50. DETAIL REQUIREMENTS.

50.1 Preliminary operations.

50.1.1 Preparation of conduction plate.

50.1.1.1 Remove any dirt and corrosion from surface of shim stock. If sufficient solder pool is already present on shim stock, proceed to step 50.1.2.

50.1.1.2 Using the soldering iron under test (UUT), bond a small pool of solder to the shim stock as shown in figure A-1.

NOTE: Heat shim stock to sufficient temperature to create a well-wetted bond between shim stock and solder.

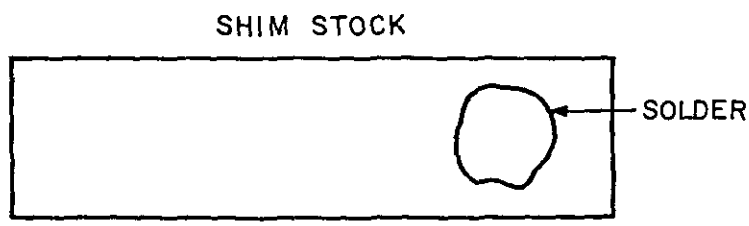


FIGURE A-1. Preparation of conduction plate.

50.1.2 Set-up of test equipment (see figure A-2).

50.1.2.1 Attach BNC cable connector to input of TRMS voltmeter (if Keithley model 132F is used, an additional banana (male) to BNC (female) adapter will be needed.

50.1.2.2 Attach ground clip probe clip to ground pin of UUT input plug.

50.1.2.3 Select appropriate range and/or function of TRMS voltmeter.

50.1.2.4 Plug TRMS voltmeter and UUT into the same duplex AC wall receptacle (Keithley model 132F is battery operated with no plug-in required).

50.1.2.5 Attach cable red wire alligator clip to conduction plate.

50.1.2.6 Attach cable black wire alligator clip to conduction plate.

50.1.2.7 Attach ground clip alligator to conduction plate.

50.2 Measuring process.

50.2.1 Measure test system RMS voltage level.

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APPENDIX A

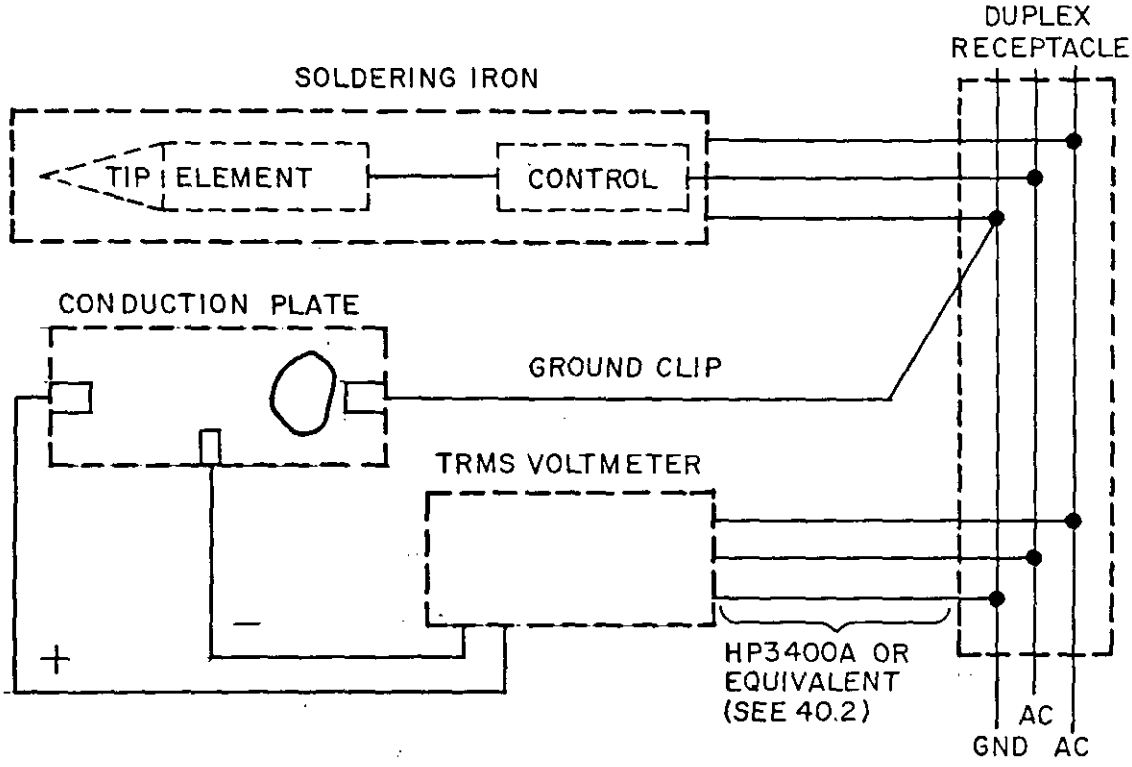


FIGURE A-2. Measurement of test system RMS voltage level.

50.2.1.1 Turn on power to TRMS voltmeter and UUT. For temperature adjustable UUTs, set temperature control to maximum temperature. Allow 15 minutes warmup.

50.2.1.2 Record TRMS voltmeter value V1. This is the test system RMS voltage level.

50.3 Measure UUT RMS voltage level (see figure A-3).

50.3.1 Remove cable black wire alligator clip and ground alligator clip from conduction plate.

50.3.2 Attach cable black wire alligator clip to ground alligator clip.

50.3.3 Remove soldering iron from holder and firmly press soldering tip against solder pool on conduction plate. Allow approximately 30 seconds for stabilization.

50.3.4 Record TRMS voltmeter value V2.

50.3.5 Calculate and record UUT RMS voltage level C1 such that C1 = V2 - V1.

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APPENDIX A

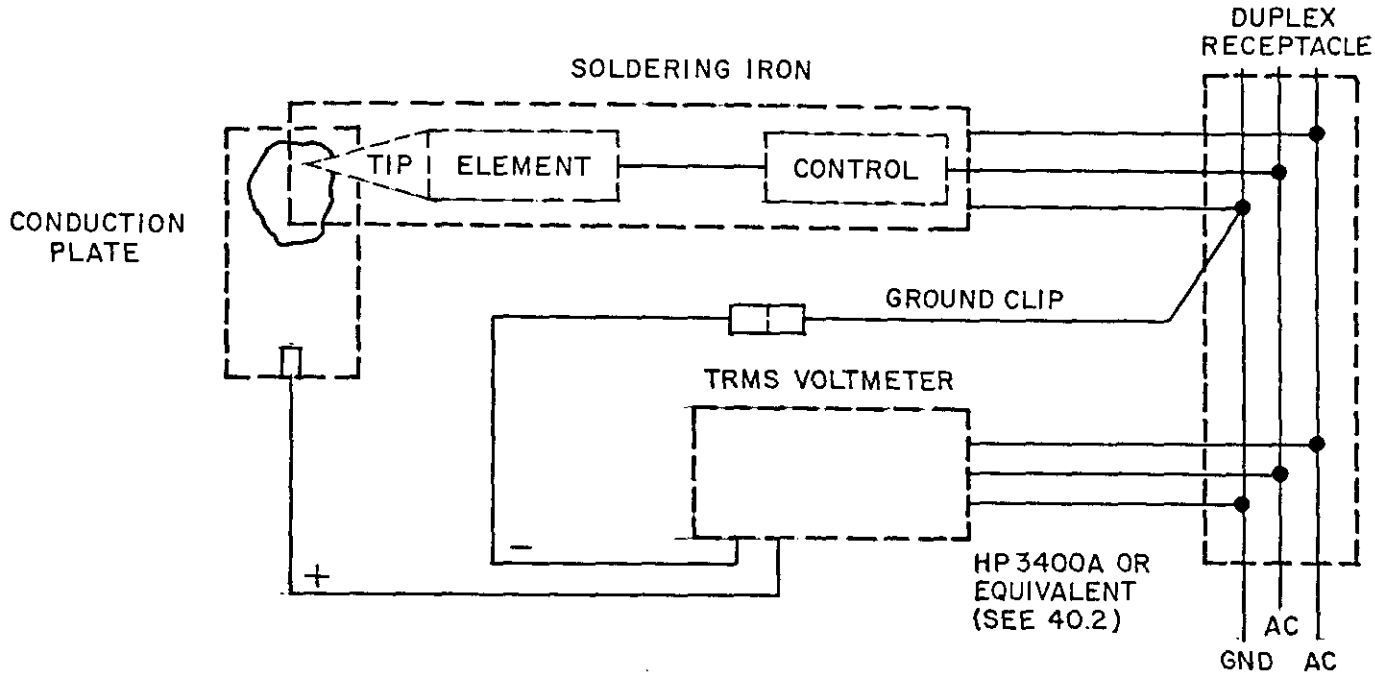


FIGURE A-3. Measurement of UUT RMS voltage level.

50.4 Measurement report. A measurement report for each UUT shall be written and shall contain, as a minimum, the following information:

a. Description of UUT (Make, Model, Identification Number).

b. Test system RMS voltage level V1.

c. RMS voltage level V2.

d. UUT RMS voltage level C1.

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APPENDIX B

STODDARD PETROLEUM DISTILLATE SOLVENT
(See 4.16.4 and 4.16.4.6)

10. SCOPE. This appendix specifies the requirements for a petroleum distillate, referred to commonly as "Stoddard solvent" which shall conform to the requirements herein and those of ASTM D 235, Type I. The requirements of this appendix correspond to the requirements of NAVAIRSYSCOM Drawing 200AS311.

10.1 Classification. The material shall be classified in accordance with ASTM D 235, Type I.

20. REFERENCED DOCUMENTS.

20.1 Government documents. The following document forms a part of this appendix to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-T-266 Thinner: Dope and Lacquer (Cellulose-Nitrate)

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Government Contracting Officer.)

20.2 Other publications. The following documents form a part of this appendix to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 235	Mineral Spirits (Petroleum Spirits) (Hydrocarbon Dry Cleaning Solvent)
ASTM E 168	Recommended Practices for General Techniques of Infrared Quantitative Analysis)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

30. DEFINITIONS.

30.1 Petroleum distillate. Petroleum distillate shall mean a hydrocarbon distilled from petroleum and not contaminated with oxygenated, halogenated or other functional solvents not found naturally in petroleum.

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APPENDIX B

40. GENERAL REQUIREMENTS.

40.1 Tabulated requirements. See table B-I.

TABLE B-I. Tabulated requirements.

Constituent	Percent; Max. by Vol.
1. Compounds with olefinic or cyclo-olefinic unsaturation	5
2. Aromatic compounds with 8 or more carbon atoms except ethylbenzene	8
3. Total of aromatics and olefins	20
4. Oxygenated solvents	Negative test

50. TEST REQUIREMENTS.

50.1 FED Spec TT-T-266, para 4.3.12 for compliance with tabulated requirements (see table B-I).

50.2 Solvent must meet all requirements of ASTM D 235.

50.3 Infrared analysis (see ASTM E 168) for compliance with 30.1.

50.4 Inspection of solvent shall be performed by either source or user as specified by procurement documents.

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APPENDIX C

DEFECT DISPOSITION FLOW CHART

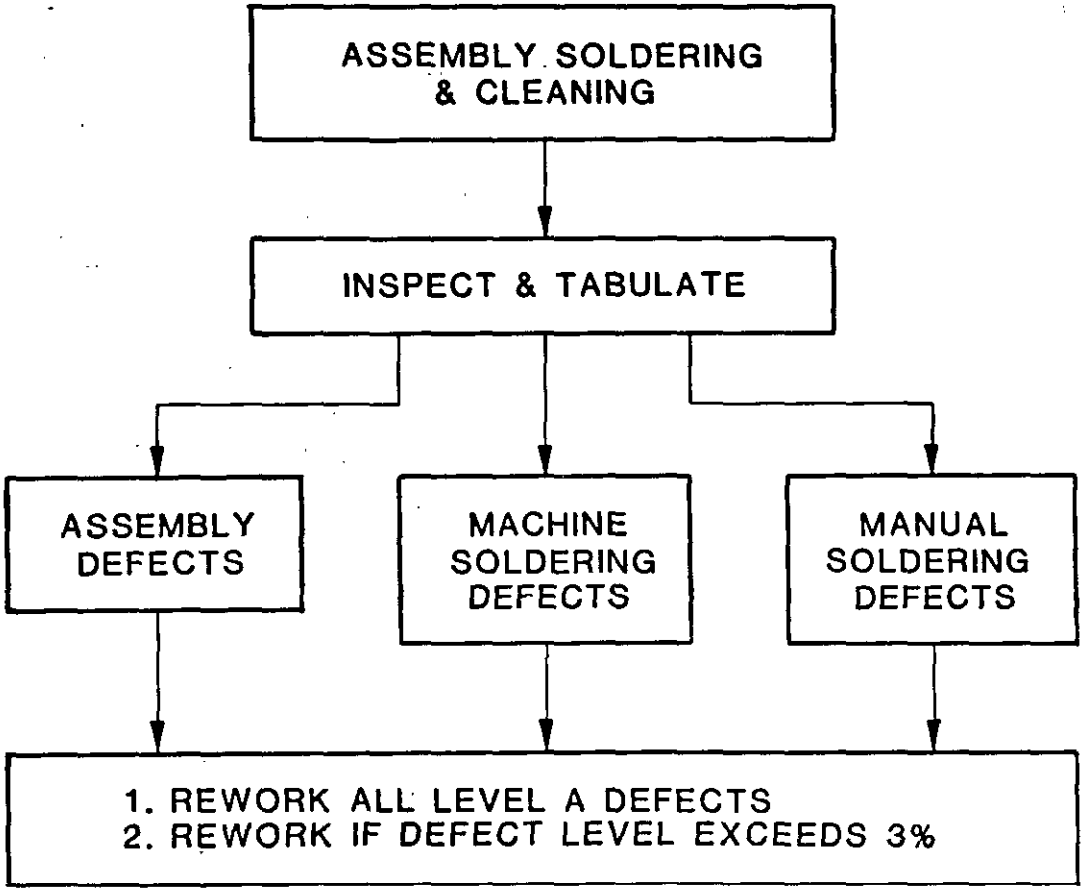


FIGURE C-1. Assembly soldering and cleaning flow chart.

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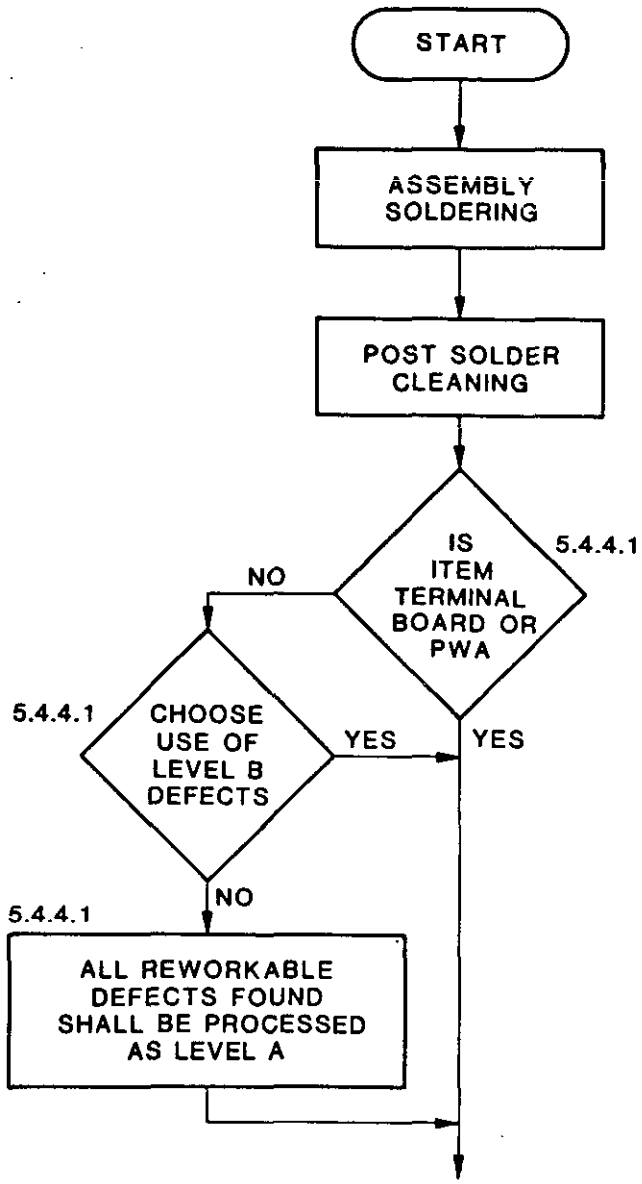


FIGURE C-2. Defect disposition flow chart.

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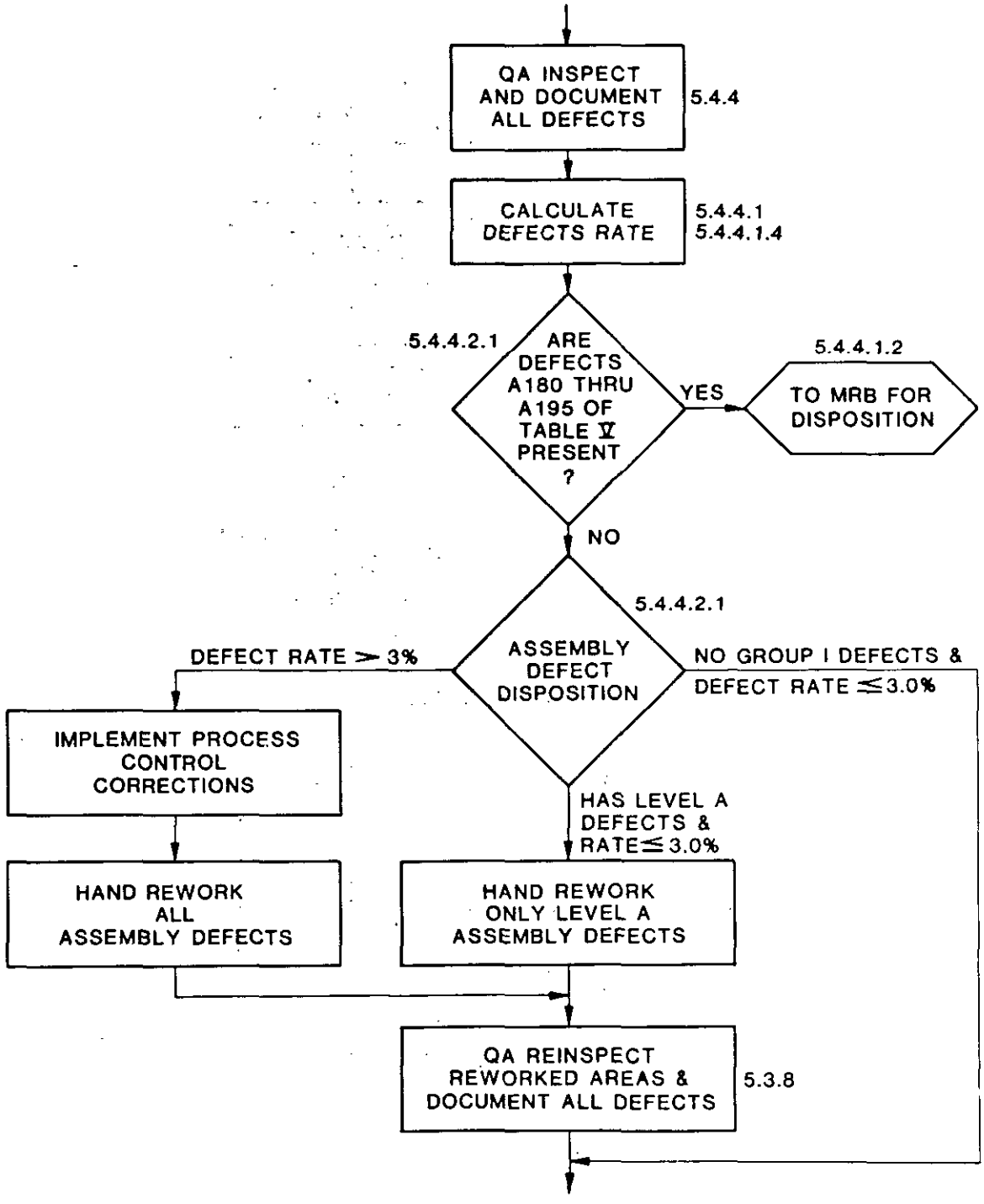


FIGURE C-2. Defect disposition flow chart (continued).

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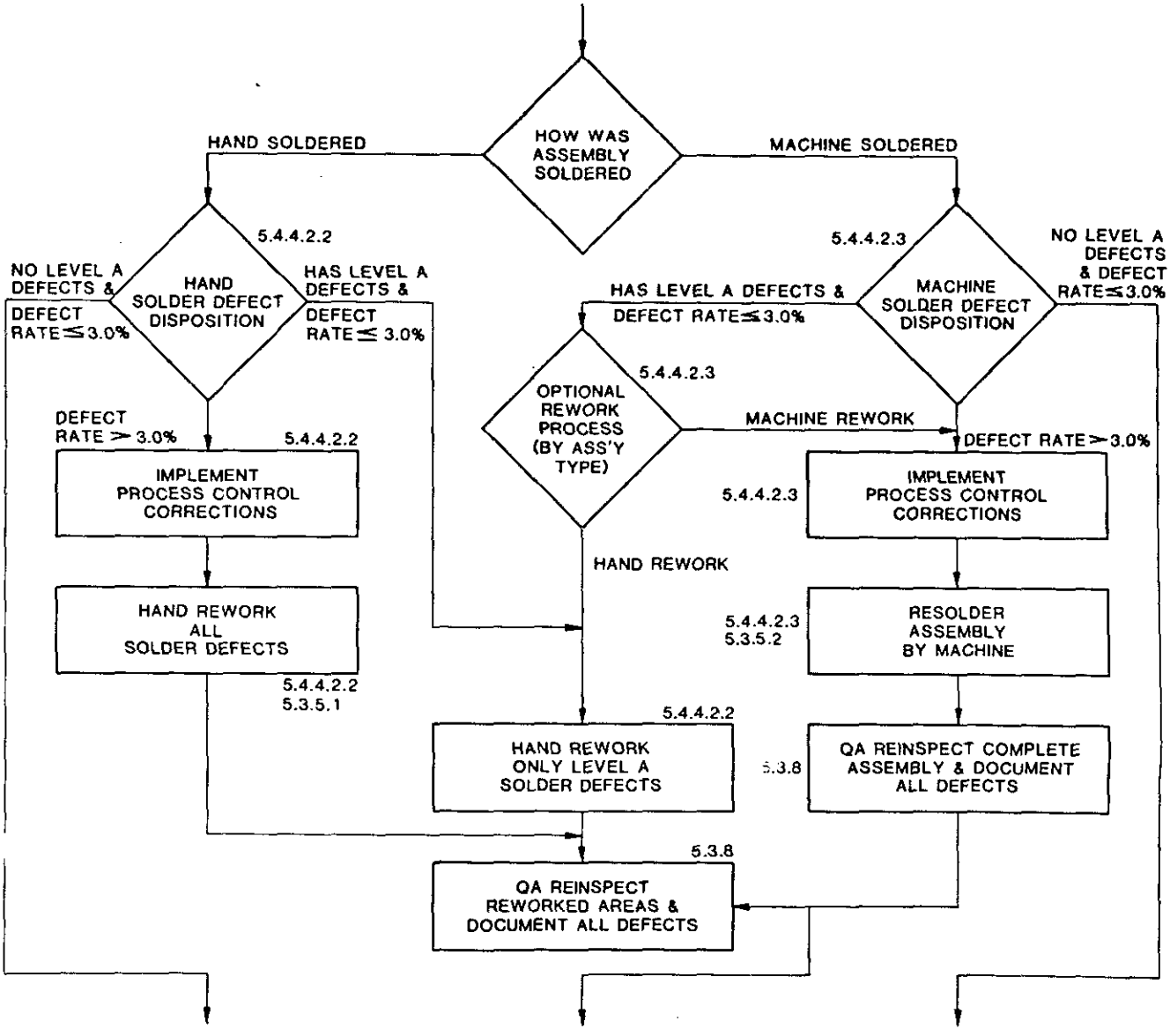


FIGURE C-2. Defect disposition flow chart (continued).

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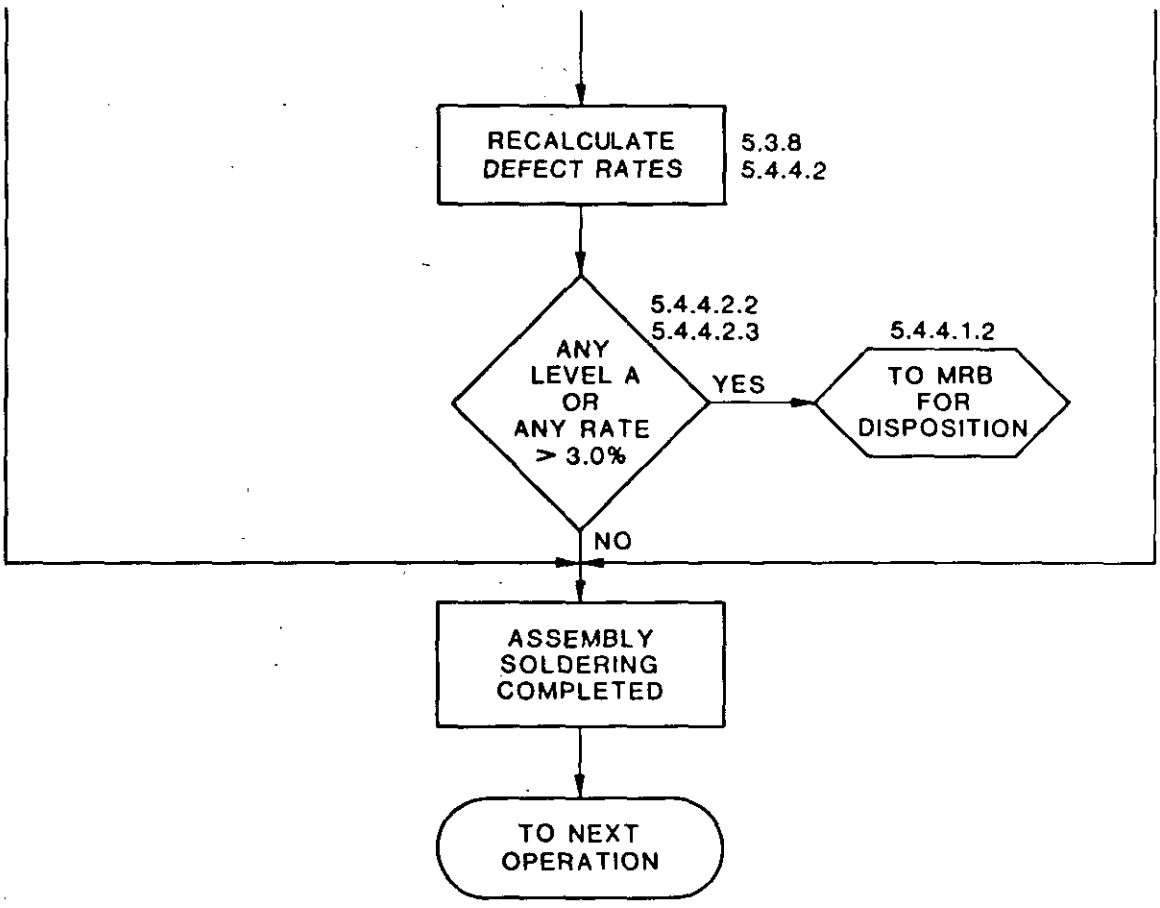


FIGURE C-2. Defect disposition flow chart (continued).

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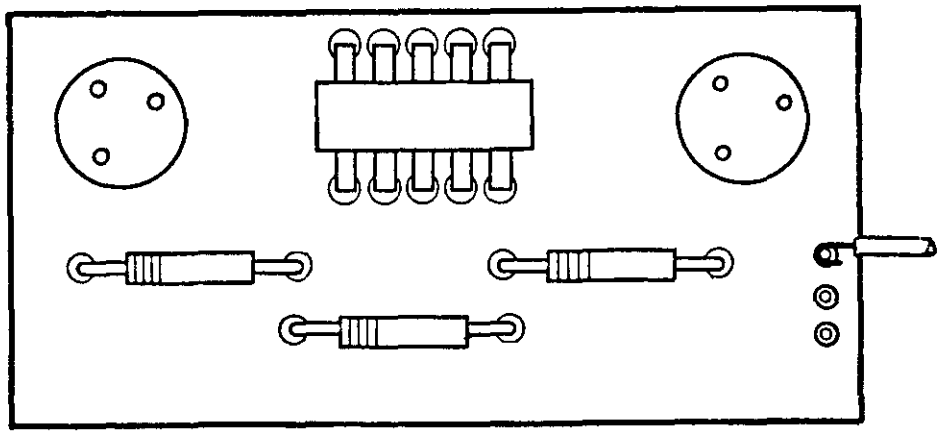
FIGURE C-3. Calculation of assembly defect rate (see 5.4.4.1.4).

Components	6
Component Leads	22
Terminals	3
Wires	1
PWB	+1
	33 = Normalizing Number

$$\frac{\text{Number of Assembly Defects observed}}{\text{Normalizing Number}} = \text{Assembly Defect Rate}$$

Sample: If there were 3 Assy Defects found on PWB below then

$\frac{3}{33}$ = 9.1% Assembly Defect Rate



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APPENDIX C

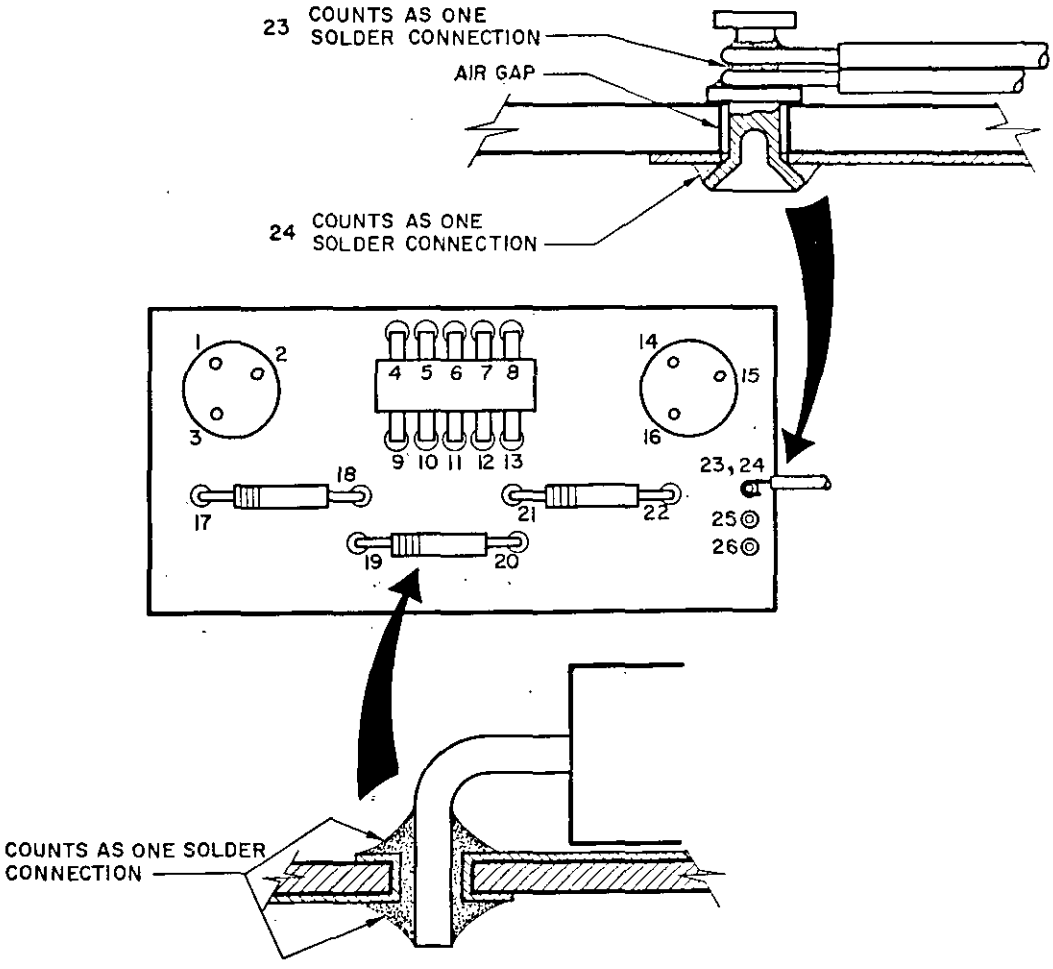
FIGURE C-4. Calculation of solder defect rates (see 5.4.4.1.4).

Solder connections = 26 = Normalizing Number

$$\frac{\text{Solder Defects Observed}}{\text{Normalizing Number}}$$

Sample: If 2 Solder Defects were found on the PWB below then,

$\frac{2}{26}$ = 7.7% Solder Defect Rate



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